

Learning Decision Boundaries for Multidimensional Anomaly Detection

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Abstract—In this article, we attempt to study a problem of identifying outliers from multiple dimensions, terming it as multidimensional anomaly detection. This task assumes that each sample corresponds to heterogeneous discriminative spaces, where each space characterizes distinct semantic information along one dimension. Consequently, the abnormalities exhibited by a sample will vary under these semantically different dimensions. In contrast to traditional anomaly detection, multidimensional anomaly detection offers a more holistic assessment of a sample's abnormalities. However, the heterogeneity of discriminative spaces leads to incomparability of the outputs from different dimensions which is the major difficulty in designing multidimensional anomaly detection methods. This article introduces a novel model, maximum margin multidimensional anomaly detection (ALOE), specifically tailored for multidimensional anomaly detection. ALOE constructs a convex optimization problem with nonlinear constraints. The primary objective is to simultaneously learn multiple decision boundaries, utilizing the maximum margin principle and covariance regularization, while distinguishing between outliers and normal samples under multiple dimensions by capturing the correlation among multiple dimensions. To obtain the optimal decision boundary under each dimension, we devise an alternating optimization method for this convex optimization problem. To validate the effectiveness of ALOE, we conduct extensive experiments on 12 real-world datasets, comparing its performance against 34 anomaly detection methods. The experimental results demonstrate the superior performance of ALOE.

Index Terms—Decision boundary, heterogeneous discriminative spaces, multidimensional anomaly detection.

I. INTRODUCTION

ANOMALY detection aims to identify outliers in a dataset that deviate significantly from normal or expected behavior [1]. In this task, we focus on identifying which samples are anomalous. However, in certain domain-specific scenarios, for the same sample, its abnormalities can present diverse results when observed from different dimensions, especially when the semantic information contained in these dimensions also plays a decisive influence on the anomalous state of the sample. This means that each sample is associated with a set of states, and each element in this set of states represents normal or

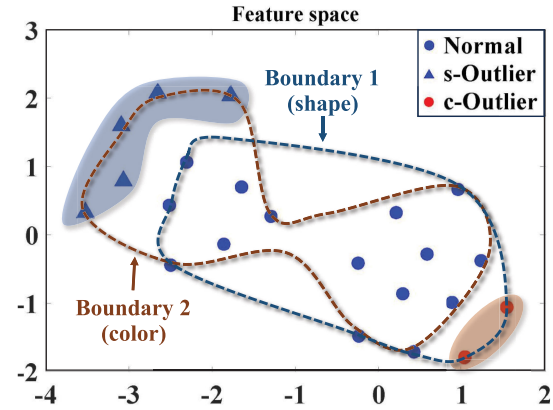


Fig. 1. Different boundaries for s-Outliers and c-Outliers (2-D distribution is obtained by t-SNE).

abnormal state for samples under one dimension. This situation exists in several application scenarios, such as water pollution detection [2], medical diagnosis [3], and fraud detection [4].

To comprehensively characterize abnormalities of samples under multiple dimensions, a reasonable approach is to give the state (normal or abnormal) of each sample under each dimension simultaneously, i.e., to assign a discriminative vector to each sample. This introduces a new task, multidimensional anomaly detection. In contrast to traditional anomaly detection, in multidimensional anomaly detection, each sample is represented by a feature vector and associated with a discriminative vector. Each element in this discriminative vector corresponds to a specific discriminative space that characterizes the state (normal or abnormal) of the sample under each dimension with the specific semantic information. This means that for the same sample, there is a diversity of semantics among the dimensions, leading to differences in the definition of anomalies. This situation poses a challenge for multidimensional anomaly detection, i.e., we need a specific decision boundary under each dimension to distinguish between normal samples and outliers. Let us take the real-world dataset “WQplants” [2] as an example.

Example 1: We randomly select 22 samples with 16 attributes¹ and two dimensions (D1: *Sphaerotilus natans* and D2: *Navicula sp.*). We use different shapes (● and ▲) to represent normal and outlier in D1 and different colors (blue and red) to represent normal and outlier in D2. As shown in Fig. 1, the s-Outliers “▲” under D1 can be

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¹These attributes are water temperature, pH, electrical conductivity, O₂, oxygen saturation, CO₂ concentration, total hardness, NO₂, NO₃, NH₄, PO₄, Cl, SiO₂, KMnO₄, K₂Cr₂O₇, and biological oxygen demand (BOD).

identified by boundary 1, while the c-Outliers “•” under D2 are separated by boundary 2. Therefore, we need to find the specific boundaries for identifying outliers under different dimensions.

The intuitive strategy for tackling the above issue involves breaking multidimensional anomaly detection down into multiple standard anomaly detection tasks and then applying standard anomaly detection methods to identify outliers. However, this strategy neglects the interdependencies among multiple dimensions, potentially undermining the generalization performance of the anomaly detection methods. To address this issue, this article introduces a novel model, termed maximum margin multidimensional anomaly detection (ALOE). Specifically, by leveraging the effectiveness of the maximum margin technique, ALOE maximizes the geometric margin between the decision boundary and the origin under each individual discriminative space, aiming to resolve the incompatibility issue arising from the modeling outputs of various dimensions. Furthermore, ALOE integrates a covariance regularization term into the maximum margin objective function, enabling it to capture the dependencies across different dimensions. Guided by this dual optimization objective, ALOE formulates a convex objective function subject to nonlinear constraints, which can be solved using alternating optimization with quadratic programming or closed-form solution in either alternating step.

The main contributions of this article are as follows.

- 1) We make the first attempt to study the multidimensional anomaly detection, and propose a novel model ALOE based on the maximum margin technique and covariance regularization for this task.
- 2) ALOE captures the relationships among multiple dimensions, which in turn improves the accuracy of identifying outliers in different dimensions, and also enhances the ALOE's generalization performance.
- 3) Extensive experiments are conducted by comparing ALOE with 34 anomaly detection methods on 12 real-world datasets for validating the effectiveness of ALOE.

The rest of the article is structured as follows. In Section II, we present a literature review on tasks that are most relevant to this article. Section III introduces our proposed method. In Section IV, we discuss the experiments with detailed insights. The conclusion is drawn in Section V.

II. RELATED WORK

A. One-Class-Based Anomaly Detection Methods

A comprehensive review of the rich literature on anomaly detection is clearly beyond the capacity of this article, so we propose one-class-based anomaly detection methods that are most relevant to this article. The rest of the popular methods are thoroughly described in a dedicated chapter in outlier analysis [5] and reviews of anomaly detection [6], [7]. One-class-based anomaly detection methods strive to learn a decision boundary that can enclose normal samples as much as possible. This category of anomaly detection methods can be further delineated into shallow and deep one-class anomaly detection methods.

1) *Shallow One-Class*: Schölkopf et al. [8] first proposed one-class support vector machine (OCSVM) which is inspired by the traditional SVM applied to detecting outliers from a dataset. Its objective is to learn an optimal hyperplane by maximizing its distance from the origin and minimizing its distance from the instance. Based on OCSVM, several studies have been proposed to improve the performance of the anomaly detector. Specifically, Song et al. [9] proposed one-class conditional random fields for sequential anomaly detection. This method can learn from the one-class dataset, and capture the temporal dependence structure. Swarnkar and Hubballi [10] proposed one-class naive bayes classifier for payload-based anomaly detection. Tian et al. [11] introduced ramp loss function into OCSVM to improve the robustness of OCSVM. Gautam et al. [12] combined multi-kernel learning with OCSVM, and proposed localized multiple kernel learning one-class anomaly detection method for improving the accuracy of outlier detection. Miao et al. [13] formulated a distributed online OCSVM for anomaly detection over networks. He et al. [14] proposed an optimal OCSVM framework for multivariate anomaly detection in telemetry.

All the above studies are carried out on OCSVM. As the research progresses, Tax and Duin [15] transformed OCSVM into another form named support vector data description (SVDD), whose purpose is to learn a minimal hypersphere that encloses as many normal samples as possible. Then, to improve the performance of SVDD, Pauwels and Ambekar [16] reanalyzed the cost function of SVDD to get a better insight into the characteristics of the solution, and introduced two new definitions of slack variables to show that one of the proposed methods behaves more robustly with respect to outliers. Wang and Cherian [17] used subspaces to label data via optimizing for two objectives jointly. Wang et al. [18] proposed an efficient adaptive multi-kernel SVDD method for outlier detection, which automatically learns the optimal weight for each training instance under different kernels using the nonnegative function. Wu and Li [19] presented a novel method for weakly supervised anomaly detection, where a limited number of labeled normal samples and a sufficient number of unlabeled samples are available for modeling.

2) *Deep One-Class*: With the popularity of deep learning, researchers have begun to further expand one-class anomaly detection into deep one-class anomaly detection. Ruff et al. [20] converted SVDD to deep via neural networks and proposed DeepSVDD to compensate for the dearth of deep learning methods for outlier detection. Erfani et al. [21] presented a hybrid model where an unsupervised deep belief network is trained to extract generic underlying features, and OCSVM is trained from the features learned by the deep belief network. Shen et al. [22] proposed the temporal hierarchical one-class network, a temporal one-class classification model for time series anomaly detection. Goyal et al. [23] designed deep robust support vector data description that is applicable to most standard domains without requiring any side-information. Chen et al. [24] introduced the interpolated Gaussian descriptor method that learns an one-class Gaussian anomaly classified trained with adversarial interpolated

training samples. Maziarka et al. [25] designed a flow-based one-class detector that finds a minimal volume bounding region for enclosing normal data. Massoli et al. [26] presented a novel framework, named multilayer one-class classification, to train and test deep learning models on the anomaly detection task. Deng and Li [27] proposed a novel teacher–student model to address the problem that using similar or identical architectures to build the teacher and student models in previous studies hinders the diversity of anomalous representations. Qiao et al. [28] defined a new form of OCSVM model based on the structure of the compressed data and the characteristics of OCSVM for addressing the problem of anomaly detection for high-dimensional sensing data. Belton et al. [29] proposed a few shot anomaly detection to accurately detect outliers having trained on few examples of the normal class. Yang et al. [30] proposed a network based on self-supervised learning and self-attentive graph convolution for effective and comprehensive anomaly detection. Chi and Mao [31] presented a robust deep SVDD-based anomaly detection algorithm built on adversarial transfer learning.

The above anomaly detection methods are used to deal with single outlying score or binary state and it is difficult to deal with multidimensional anomaly detection, due to the fact that correlations between different dimensions are not considered.

B. Multidimensional Classification

Multidimensional anomaly detection does not refer to multiple dimensions of a sample’s features, but rather to a set of states for the sample obtained from multiple dimensions. Currently, there is little research on multidimensional anomaly detection. Therefore, we briefly introduce its similar task multidimensional classification. Multidimensional classification is a new multioutput learning paradigm, where each sample in this framework is annotated with labels from multiple semantic class spaces [32].

For multidimensional classification, Read et al. [33] presented a new method based on multilabel research for multidimensional classification. Wang et al. [34] presented a novel neural network-based model to integrate the label embedding and feature augmentation techniques to learn label correlations. Jia and Zhang [35] attempted to feature manipulation in input space for multidimensional classification, and designed a simple yet effective approach to enrich the original feature space with augmented features. Then, they also proposed another two methods [36], [37] for multidimensional classification. Huang et al. [38] attempted to deal with the multidimensional classification problem under the semi-supervised learning setting, so that they developed a novel multidimensional classification approach to solve the resulting semi-supervised multidimensional classification problem. Jia and Zhang [32], [39] designed a new classification framework to investigate which models objects with rich semantics by encompassing heterogeneous label spaces and multilabel annotations. Liu et al. [40] simultaneously conducted label space fusion and considered the comprehensive label correlations to fully exploit the latent discriminative information hidden in the output space.

C. Multiclass Anomaly Detection

Multiclass anomaly detection aims to train an anomaly detector based on multiple classes’ samples to detect outliers in all classes.

For this task, You et al. [41] presented a unified framework to accomplish anomaly detection for multiple classes. Based on the probabilistic framework, several studies have been proposed. Specifically, Lu et al. [42] proposed a hierarchical vector quantized prototype-oriented Transformer. Wang Tavakkoli [43] combined the reconstruction-based structure with the latent space denoising diffusion probabilistic model. Li et al. [44] designed a vague prototype-oriented diffusion model which can extract fundamental information from the condition for preventing the same shortcut from the input layer. Based on the similar strategy, He et al. [45] developed a framework consisting of a pixel-space autoencoder, a latent-space semantic-guided network, and a feature-space pretrained feature extractor. Liu and Wang [46] designed a model based on the unified mixed-attention autoencoder to implement multiclass anomaly detection. Wu et al. [47] represented a novel iterative memory review network that selects prior reconstruction information and strengthening the interactivity between network modules. Kang et al. [48] employed an encoder–decoder architecture, and designed a masked subspace-like transformer for multiclass anomaly detection.

We further analyze the difference among anomaly detection, multiclass anomaly detection, and multidimensional anomaly detection. Specifically, anomaly detection attempts to train an anomaly detector that can detect outliers from a dataset that are significantly different from the vast majority of samples. Then, multiclass anomaly detection aims to train a generic anomaly detector that can simultaneously detect outliers from multiple classes that differ from the vast majority of samples in the corresponding class. In contrast, multidimensional anomaly detection tries to train an anomaly detector that can simultaneously detect outliers that differ from the vast majority of samples under different dimensions from the same dataset.

III. METHODOLOGY

A. Problem Definition

Suppose that we have an input space $\mathcal{X} \in \mathbb{R}^d$, and an output space $\mathcal{Y} = L_1 \times \cdots \times L_m$ which corresponds to the Cartesian product of m discriminative spaces associated with m multiple dimensions. Each space $L_j (1 \leq j \leq m)$ contains two states l_+ and l_- , where l_+ and l_- denote that the sample is normal and abnormal, respectively. Given a training set $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$, where $\mathbf{x}_i = [x_{i1}, \dots, x_{id}]^\top$ denotes a feature vector and $\mathbf{y}_i = [y_{i1}, \dots, y_{im}]^\top$ denotes the discriminative vector associated with $\mathbf{x}_i$ where each component $y_{ij} \in \{l_+, l_-\}$. Multidimensional anomaly detection aims to train an anomaly detector $f : \mathcal{X} \mapsto \mathcal{Y}$ over $\mathcal{D}$ which can assign a discriminative vector $f(\hat{\mathbf{x}}) \in \mathcal{Y}$ for a sample $\hat{\mathbf{x}}$. Each element of this vector can describe the sample’s state (normal or abnormal) in each dimension.

B. Design of ALOE

We address multidimensional anomaly detection by transforming this task into m standard anomaly detection problems. Without loss of generality, let $\mathcal{D}^j = \{(\mathbf{x}_i^j, l_+^j)\}_{i=1}^{n_j} \cup \{(\mathbf{x}_i^j, l_-^j)\}_{i=n_j+1}^n$ denote the training set with n_j normals and $n - n_j$ outliers in the j th label space. Assuming that all normals in $\mathcal{D}^j$ are separated from the origin by the hyperplane $(\mathbf{w}_j, \rho_j)$, we can define the margin of this hyperplane as $2/\|\mathbf{w}_j\|$ by appropriately normalizing $(\mathbf{w}_j, \rho_j)$. And then, the maximum margin hyperplane for $\mathcal{D}^j$ is obtained by minimizing $\|\mathbf{w}_j\|^2/2$. To improve the generalization of this hyperplane, we can add the empirical risk by introducing slack variables $\xi_i^j = [\xi_1^j, \dots, \xi_n^j]$. Under m dimensions, we define $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_m] \in \mathbb{R}^{d \times m}$, $\boldsymbol{\rho} = [\rho_1, \dots, \rho_m]^\top$, and $\boldsymbol{\xi} = [\xi^1, \dots, \xi^m]^\top \in \mathbb{R}^{mn \times 1}$. The maximum margin optimization problem for multidimensional anomaly detection is as follows:

$$\begin{aligned} \min_{\mathbf{w}, \boldsymbol{\rho}, \boldsymbol{\xi}} \quad & \frac{\lambda_1}{2} \text{tr}(\mathbf{W}\mathbf{W}^\top) - \sum_{j=1}^m \rho_j + \sum_{j=1}^m \sum_{i=1}^{n_j} \sum_{l=n_j+1}^n \xi_i^j + \xi_l^j \\ \text{s.t.} \quad & \mathbf{w}_j^\top \mathbf{x}_i^j \geq \rho_j - \xi_i^j, \quad i = 1, \dots, n_j \\ & \mathbf{w}_j^\top \mathbf{x}_l^j \leq \rho_j + \xi_l^j, \quad l = n_j + 1, \dots, n \\ & \xi_i^j, \xi_l^j \geq 0, \quad j = 1, \dots, m \end{aligned} \quad (1)$$

where λ_1 is the trade-off parameter and $\text{tr}(\cdot)$ denotes the trace of a square matrix. We should note that the optimization problem (1) only considers the abnormal condition under each dimension independently while ignores the potential dependence among different dimensions. Inspired by the covariance regularization strategy for dependency modeling, we introduce the column covariance matrix ($\mathbf{C} \in \mathbb{R}^{m \times m}$) of $\mathbf{W}$. Then, the optimization problem (1) is rewritten as

$$\begin{aligned} \min_{\mathbf{w}, \boldsymbol{\rho}, \boldsymbol{\xi}, \mathbf{C}} \quad & \frac{\lambda_1}{2} \text{tr}(\mathbf{W}\mathbf{W}^\top) + \frac{\lambda_2}{2} \text{tr}(\mathbf{W}\mathbf{C}^{-1}\mathbf{W}^\top) - \sum_{j=1}^m \rho_j \\ & + \sum_{j=1}^m \sum_{i=1}^{n_j} \xi_i^j + \sum_{j=1}^m \sum_{l=n_j+1}^n \xi_l^j \\ \text{s.t.} \quad & \mathbf{w}_j^\top \mathbf{x}_i^j \geq \rho_j - \xi_i^j, \quad i = 1, \dots, n_j \\ & \mathbf{w}_j^\top \mathbf{x}_l^j \leq \rho_j + \xi_l^j, \quad l = n_j + 1, \dots, n \\ & \mathbf{C} \succeq 0, \quad \text{tr}(\mathbf{C}) \leq 1, \quad \xi_i^j, \xi_l^j \geq 0, \quad j = 1, \dots, m \end{aligned} \quad (2)$$

where λ_2 is the trade-off parameter. $\mathbf{C} \succeq 0$ ensures that the covariance matrix is positive semi-definite and $\text{tr}(\mathbf{C}) \leq 1$ penalizes its complexity. Minimizing the second term aims to maximum a posteriori estimation of $\mathbf{W}$ with assuming the matrix-variate normal distribution $\mathbf{W} \sim \mathcal{MN}_{d \times m}(\mathbf{W}|\mathbf{0}_{d \times m}, \mathbf{I}_{d \times d} \otimes \mathbf{C})$ over it, where $\mathbf{0}_{d \times m}$ denotes the zero matrix and $\mathbf{I}_{d \times d}$ is identity matrix which denotes that the features are assumed uncorrelated with each other. The corresponding probability density function is defined as $p(\mathbf{W}|\mathbf{0}_{d \times m}, \mathbf{I}_{d \times d} \otimes \mathbf{C}) = ((\exp(-(1/2)\text{tr}(\mathbf{I}_{d \times d}^{-1}(\mathbf{W} - \mathbf{0}_{d \times m})\mathbf{C}^{-1}(\mathbf{W} - \mathbf{0}_{d \times m})^\top)))/((2\pi)^{md/2}|\mathbf{I}_{d \times d}|^{m/2}|\mathbf{C}|^{d/2}))$.

For covariance matrix, each element represents the covariance between two dimensions, which reflects the positive or negative relationships of two dimensions. Using covariance

regularization makes the covariance matrix more numerically stable by restricting certain elements of the covariance matrix. At the same time, regularization does not destroy the correlation information in the covariance matrix, and it fine-tunes the matrix to make it more realistic while maintaining the original correlation structure.

C. ALOE Optimization

The objective function of the optimization problem (2) is convex except for the second term. For the second term, it is reformulated as the summation of d items $\text{tr}(\mathbf{W}\mathbf{C}^{-1}\mathbf{W}^\top) = \sum_{i=1}^d \mathbf{W}_i \mathbf{C}^{-1} \mathbf{W}_i^\top$, where $\mathbf{W}_i$ is the i th row of $\mathbf{W}$. Based on the convexity of $\mathbf{W}_i \mathbf{C}^{-1} \mathbf{W}_i^\top$, when $\mathbf{C}$ is positive semi-definite, $\text{tr}(\mathbf{W}\mathbf{C}^{-1}\mathbf{W}^\top)$ is also convex after summation operation. Therefore, the second term has the convexity, so that the objective function of the optimization problem (2) is convex. We design an alternating procedure strategy to solving this optimization problem. Specifically, we first optimize $\mathbf{W}$ and $\boldsymbol{\rho}$ with fixed $\mathbf{C}$, and then optimize $\mathbf{C}$ with fixed $\mathbf{W}$, $\boldsymbol{\rho}$.

1) *Optimizing $\mathbf{W}$ and $\boldsymbol{\rho}$ With $\mathbf{C}$ Fixed:* If we fix the $\mathbf{C}$, the optimization problem (2) is transformed as follows:

$$\begin{aligned} \min_{\mathbf{w}, \boldsymbol{\rho}, \boldsymbol{\xi}} \quad & \frac{\lambda_1}{2} \text{tr}(\mathbf{W}\mathbf{W}^\top) + \frac{\lambda_2}{2} \text{tr}(\mathbf{W}\mathbf{C}^{-1}\mathbf{W}^\top) \\ & - \sum_{j=1}^m \rho_j + \sum_{j=1}^m \sum_{i=1}^{n_j} \xi_i^j + \sum_{j=1}^m \sum_{l=n_j+1}^n \xi_l^j \\ \text{s.t.} \quad & \mathbf{w}_j^\top \mathbf{x}_i^j \geq \rho_j - \xi_i^j, \quad i = 1, \dots, n_j \\ & \mathbf{w}_j^\top \mathbf{x}_l^j \leq \rho_j + \xi_l^j, \quad l = n_j + 1, \dots, n \\ & \xi_i^j, \xi_l^j \geq 0, \quad j = 1, \dots, m. \end{aligned} \quad (3)$$

To optimize $\mathbf{W}$, $\boldsymbol{\rho}$, and $\boldsymbol{\xi}$, given Lagrange multipliers $\boldsymbol{\alpha} = [\alpha_1^1, \dots, \alpha_n^1, \dots, \alpha_1^m, \dots, \alpha_n^m]^\top \in \mathbb{R}^{mn \times 1}$ and $\boldsymbol{\beta} = [\beta_1^1, \dots, \beta_n^1, \dots, \beta_1^m, \dots, \beta_n^m]^\top \in \mathbb{R}^{mn \times 1}$, the Lagrange function of the optimization problem (3) is

$$\begin{aligned} \mathcal{L} = \quad & \frac{\lambda_1}{2} \text{tr}(\mathbf{W}\mathbf{W}^\top) + \frac{\lambda_2}{2} \text{tr}(\mathbf{W}\mathbf{C}^{-1}\mathbf{W}^\top) - \sum_{j=1}^m \rho_j \\ & + \sum_{j=1}^m \sum_{i=1}^{n_j} \xi_i^j + \sum_{j=1}^m \sum_{l=n_j+1}^n \xi_l^j - \sum_{j=1}^m \sum_{i=1}^{n_j} \beta_i^j \xi_i^j \\ & - \sum_{j=1}^m \sum_{i=1}^{n_j} \alpha_i^j (\mathbf{w}_j^\top \mathbf{x}_i^j - \rho_j + \xi_i^j) - \sum_{j=1}^m \sum_{l=n_j+1}^n \beta_l^j \xi_l^j \\ & + \sum_{j=1}^m \sum_{l=n_j+1}^n \alpha_l^j (\mathbf{w}_j^\top \mathbf{x}_l^j - \rho_j - \xi_l^j). \end{aligned} \quad (4)$$

We take the partial derivatives of $\mathbf{W}$, $\boldsymbol{\rho}$, and $\boldsymbol{\xi}$ for the Lagrangian function, respectively. The results are listed in Appendix. With introducing partial derivatives into Lagrange function (4), the terms associated with ρ_j , ξ_i^j , ξ_l^j , β_i^j , and β_l^j can be eliminated. Lagrange function (4) is simplified as

$$\begin{aligned} \mathcal{L}(\mathbf{W}; \boldsymbol{\alpha}) = \quad & \frac{1}{2} \text{tr}(\mathbf{W}(\lambda_1 \mathbf{I}_m + \lambda_1 \mathbf{C}^{-1})\mathbf{W}^\top) \\ & - \sum_{j=1}^m \sum_{i=1}^{n_j} \sum_{l=n_j+1}^n (\alpha_i^j \mathbf{w}_j^\top \mathbf{x}_i^j + \alpha_l^j \mathbf{w}_j^\top \mathbf{x}_l^j) \\ & + \alpha_i^j + \alpha_l^j. \end{aligned} \quad (5)$$

Let $\mathbf{H} = (\lambda_1 \mathbf{C} + \lambda_2 \mathbf{I}_m)^{-\top} \mathbf{C}^\top$, and $\mathbf{H}_{j_1 j_2} = \mathbf{e}_{j_1}^\top \mathbf{H} \mathbf{e}_{j_2}$. The Lagrange function (5) is further transformed as

$$\begin{aligned} \mathcal{L}(\alpha) = & -\frac{1}{2} \sum_{j_1, j_2=1}^m \sum_{i_1=1}^{n_{j_1}} \sum_{i_2=1}^{n_{j_2}} \sum_{l_1=n_{j_1}+1}^n \sum_{l_2=n_{j_2}+1}^n \alpha_{i_1}^{j_1} \alpha_{i_2}^{j_2} \mathbf{H}_{j_1 j_2} \mathbf{x}_{i_1}^{j_1 \top} \mathbf{x}_{i_2}^{j_2} \\ & + \alpha_{l_1}^{j_1} \alpha_{l_2}^{j_2} \mathbf{H}_{j_1 j_2} \mathbf{x}_{l_1}^{j_1 \top} \mathbf{x}_{l_2}^{j_2} + \sum_{j=1}^m \sum_{i=1}^{n_j} \sum_{l=n_j+1}^n \alpha_i^j + \alpha_l^j. \end{aligned} \quad (6)$$

Then, the dual problem of optimization problem (3) is as follows:

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2} \sum_{j_1=1}^m \sum_{j_2=1}^m \sum_{i_1=1}^{n_{j_1}} \sum_{i_2=1}^{n_{j_2}} \sum_{l_1=n_{j_1}+1}^n \sum_{l_2=n_{j_2}+1}^n \alpha_{i_1}^{j_1} \alpha_{i_2}^{j_2} \mathbf{H}_{j_1 j_2} \mathbf{x}_{i_1}^{j_1 \top} \mathbf{x}_{i_2}^{j_2} \\ & + \alpha_{l_1}^{j_1} \alpha_{l_2}^{j_2} \mathbf{H}_{j_1 j_2} \mathbf{x}_{l_1}^{j_1 \top} \mathbf{x}_{l_2}^{j_2} \\ \text{s.t.} \quad & \sum_{i=1}^{n_j} \alpha_i^j + \sum_{l=n_j+1}^n \alpha_l^j = 1, \quad 0 \leq \alpha_i^j \leq 1, \quad 0 \leq \alpha_l^j \leq 1. \end{aligned} \quad (7)$$

This dual problem is a quadratic programming problem with m equality constraints, which is solved by the quadratic programming solver.

2) *Optimizing C With W and ρ Fixed:* When we fix $\mathbf{W}$ and ρ , the optimization problem (2) is written as follows:

$$\min_{\mathbf{C}} \text{tr}(\mathbf{W} \mathbf{C}^{-1} \mathbf{W}^\top), \quad \text{s.t. } \mathbf{C} \succeq 0, \quad \text{tr}(\mathbf{C}) \leq 1. \quad (8)$$

Based on the property of the matrix trace calculation, and $\text{tr}(\mathbf{C}) \leq 1$, we have

$$\begin{aligned} \text{tr}(\mathbf{W} \mathbf{C}^{-1} \mathbf{W}^\top) &= \text{tr}(\mathbf{C}^{-1} \mathbf{W}^\top \mathbf{W}) \\ &\geq \text{tr}(\mathbf{C}^{-1} \mathbf{W}^\top \mathbf{W}) \text{tr}(\mathbf{C}) \\ &\geq \text{tr}\left(\mathbf{C}^{-\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} \mathbf{C}^{-\frac{1}{2}}\right) \text{tr}\left(\mathbf{C}^{\frac{1}{2}} \mathbf{C}^{\frac{1}{2}}\right). \end{aligned}$$

Since $\mathbf{W}^\top \mathbf{W}$ and $\mathbf{C}$ are symmetric matrices, we have the following lemma.

Lemma 1: Given $(\mathbf{W}^\top \mathbf{W})^{(1/2)} \mathbf{C}^{-(1/2)}, \mathbf{C}^{(1/2)} \in \mathbb{R}^{m \times m}$, we have the following inequality:

$$\begin{aligned} \text{tr}\left(\mathbf{C}^{-\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} \mathbf{C}^{-\frac{1}{2}}\right) \text{tr}\left(\mathbf{C}^{\frac{1}{2}} \mathbf{C}^{\frac{1}{2}}\right) \\ \geq \left(\text{tr}\left(\mathbf{C}^{-\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} \mathbf{C}^{\frac{1}{2}}\right)\right)^2. \end{aligned}$$

Only when $(\mathbf{W}^\top \mathbf{W})^{(1/2)} \mathbf{C}^{-(1/2)} = \mu \cdot \mathbf{C}^{(1/2)}$, we have

$$\begin{aligned} \text{tr}\left(\mathbf{C}^{-\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} \mathbf{C}^{-\frac{1}{2}}\right) \text{tr}\left(\mathbf{C}^{\frac{1}{2}} \mathbf{C}^{\frac{1}{2}}\right) \\ = \left(\text{tr}\left(\mathbf{C}^{-\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} \mathbf{C}^{\frac{1}{2}}\right)\right)^2. \end{aligned}$$

The proof is listed in Appendix. Based on Lemma 1, we can get the lower bound of the objective function in optimization problem (8)

$$\begin{aligned} \text{tr}(\mathbf{W} \mathbf{C}^{-1} \mathbf{W}^\top) &\geq \text{tr}\left(\mathbf{C}^{-\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} \mathbf{C}^{-\frac{1}{2}}\right) \text{tr}\left(\mathbf{C}^{\frac{1}{2}} \mathbf{C}^{\frac{1}{2}}\right) \\ &\geq \left(\text{tr}\left(\mathbf{C}^{-\frac{1}{2}} (\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}} \mathbf{C}^{\frac{1}{2}}\right)\right)^2 \\ &\geq \left(\text{tr}\left((\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}}\right)\right)^2. \end{aligned} \quad (9)$$

Algorithm 1 ALOE

Require: Training set $\mathcal{D}$, trade-off parameters λ_1 and λ_2 , test instance $\hat{\mathbf{x}}$

Ensure: The discriminative vector $\hat{\mathbf{y}}$ for $\hat{\mathbf{x}}$.

- 1: Transform the training set $\mathcal{D}$ into m training subsets for each dimension;
 - 2: Initialize $\mathbf{C} = \frac{1}{m} \mathbf{I}_m$ and $\alpha = \mathbf{0}$;
 - 3: **repeat**
 - 4: Solve the optimization problem (7);
 - 5: Calculate $\mathbf{C}$ according to Equation (10);
 - 6: **until** Convergence
 - 7: **return** $\hat{\mathbf{y}}$ according to $\hat{\mathbf{y}} = \text{sign}(\mathbf{W}^\top \hat{\mathbf{x}} - \rho)$
-

With inequality (9), when both $(\mathbf{W}^\top \mathbf{W})^{(1/2)} \mathbf{C}^{-(1/2)} = \mu \mathbf{C}^{(1/2)}$ and $\text{tr}(\mathbf{C}) = 1$ hold, $\text{tr}(\mathbf{W} \mathbf{C}^{-1} \mathbf{W}^\top)$ reaches its minimum value $(\text{tr}((\mathbf{W}^\top \mathbf{W})^{(1/2)}))^2$. Then, the closed-form solution for $\mathbf{C}$ is

$$\mathbf{C} = \frac{(\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}}}{\text{tr}((\mathbf{W}^\top \mathbf{W})^{\frac{1}{2}})} \quad (10)$$

where $\mathbf{W}^\top \mathbf{W} = \sum_{j_1=1}^m \sum_{j_2=1}^m \sum_{i_1=1}^{n_{j_1}} \sum_{i_2=1}^{n_{j_2}} \sum_{l_1=n_{j_1}+1}^n \sum_{l_2=n_{j_2}+1}^n \alpha_{i_1}^{j_1} \alpha_{i_2}^{j_2} \mathbf{H}_{j_1 j_2} \mathbf{H}_{j_1 j_2}^\top \langle \mathbf{x}_{i_1}^{j_1}, \mathbf{x}_{i_2}^{j_2} \rangle + \alpha_{l_1}^{j_1} \alpha_{l_2}^{j_2} \mathbf{H}_{j_1 j_2} \mathbf{H}_{j_1 j_2}^\top \langle \mathbf{x}_{l_1}^{j_1}, \mathbf{x}_{l_2}^{j_2} \rangle$.

The optimization objective is a convex function with the following preferred properties. Specifically, a convex function will not have a function value between any two points in the domain of definition that is lower than the value on the line connecting these two points. This property ensures that the function has only one optimal solution globally, and there will be no local optimal solution interfering with the solution of the global optimal solution. This is why we use the alternating optimization with quadratic programming or closed-form solution in either alternating step to obtain the global optimal solution. The quadratic programming solution is used when we solve for $\mathbf{W}$ and ρ . This algorithm has been shown to converge. When we solve $\mathbf{C}$, this article also shows that the solution has a lower bound and the convergence of the optimization algorithm. Thus, the solution algorithm is guaranteed to be convergent regardless of the order.

As the two alternating optimizing steps converge, we obtain the predictive model. Then, given a test set $\mathcal{T} = \{(\hat{\mathbf{x}}_i, \mathbf{y}_i)\}_{i=1}^p$, the discriminative vector for a test instance $\hat{\mathbf{x}}_i$ can be predicted by $\hat{\mathbf{y}}_i = \text{sign}(\mathbf{W}^\top \hat{\mathbf{x}}_i - \rho)$, where $\text{sign}(\cdot)$ denotes the (element-wise) signed function. The key to ALOE being able to address multidimensional anomaly detection is to learn the specific decision boundary under each dimension simultaneously using maximum margin technique and covariance regularization, thus ensuring that ALOE can collaboratively determine the state of each sample under multiple dimensions.

When we fix $\mathbf{W}$ and ρ , the optimization problem (2) can also be reformulated as the optimization problem (8). The procedure of the ALOE is shown in Algorithm 1. We further analyze the computational complexity. The quadratic programming problem (7) with m variables can be solved with $\mathcal{O}(m^3)$ time complexity. The square root for one matrix in (10) with $m \times m$ elements are obtained with $\mathcal{O}(m^3)$. ALOE has

computational complexity of $\mathcal{O}(T \cdot m \cdot N^3 + T \cdot m^3)$, where T is the number of alternating rounds and N is the worst complexity of solving the optimization problem (7). The kernel version of ALOE is shown in Appendix.

IV. EXPERIMENTS AND RESULTS

We conduct experiments on a set of 12 real-world datasets, aiming to demonstrate the superior performance of ALOE. All experiments are conducted on a PC with AMD 3990X 64-Core Processor 2.90 GHz CPU, and 256 GB memory, running Windows10 Professional. The algorithms are implemented in MATLAB (R2023b).

A. Experimental Settings

1) *Datasets*: We select 12 multidimensional datasets and reconstruct them to ensure applicability to our task. Specifically, *Edm* [49] is a dataset for the task of predicting control operations during electrical discharge machining process, where the two dimensions correspond to two controlling variables gap and flow, respectively. Each control variable has three actions including increase the control variable, no action, and decrease the control variable. In multidimensional anomaly detection, for each control variable, we define the increase the control variable as normal samples, and no action and decrease the control variable are defined as outliers. *Jura* [50] is a dataset for the task of predicting the concentration of several metal pollutants in a region of the Swiss Jura. The two dimensions correspond to land and rock, respectively. In multidimensional anomaly detection, for each dimension, we define rocks containing the primary variable cadmium as normal samples and rocks containing the secondary variables nickel and zinc as outliers. *Enb* [51] is a dataset for predicting some building parameters of residential buildings. The two dimensions correspond to overall height and glazing area, respectively. For multidimensional anomaly detection, in overall height, we define heights of 3.5 as normal samples, and heights of 7 as outliers. In glazing area, we define those with glazing as normal samples and those without glazing as outliers. *Flare1*² is a dataset for predicting the number of times certain types of solar flare occurred within 24 h period, where the three dimensions correspond to common, moderate, and severe solar flares, respectively. In multidimensional anomaly detection, among three types of solar flares, we define no solar flare occurrence as normal samples and the others as outliers. *WQanimals* and *WQplants* [2] are two datasets for predicting animal and plant species in Slovenian rivers, respectively. Each of them has seven dimensions correspond to relative representation of different species. For multidimensional anomaly detection, in each species, we define a sampling site in the river is normal when the species is present in the river, and an outlier if the species is not present in this river. *Voice* [52] is a dataset for predicting some characteristics of a piece of human voice from different dimensions containing mean frequency range and speaker's gender. For multidimensional anomaly detection, in the first dimension, we define the

TABLE I
DESCRIPTION OF REAL-WORLD DATASETS

Dataset	#Sample	#Dimension	#Outlier in each dimension
Edm	154	2	26/68
Jura	359	2	141/235
Enb	768	2	384/240
Flare1	323	3	36/32/7
WQanimals	1060	7	299/620/343/465/334/389/311
WQplants	1060	7	456/283/305/218/500/592/262
Voice	3136	2	1400/1562
Song	785	3	436/160/157
BeLaE	1930	5	684/633/815/648/637
COIL2000	9822	5	4668/2694/4402/3919/586
TIC2000	9822	3	4668/3919/586
Pain	9734	10	213/1109/667/87/102/1376/136/477/419/491

human's mean frequency in [80, 120 Hz] as normal, and over 120 Hz as outliers. In the second dimension, we define the male as normal samples, and female as outliers. *Song* [35] is a dataset for predicting properties of 785 songs from three dimensions including song's emotion, genre, and scenarios. For multidimensional anomaly detection, on the song emotion dimension, we define happy emotions as normal samples, and sad and cathartic emotions as outliers. On the song genre dimension, we define popular songs as normal samples and others as outliers. On the song scenarios dimension, we define common scenes as normal samples, and wedding and nightclub scenes as outliers. *BeLaE* [53] is a dataset for predicting students' answers to five questions in a questionnaire. These questions are on the importance of certain properties of their future jobs. For multidimensional anomaly detection, on each question, we define answers getting grade 5 (very important) as normal samples, and answers with other grades as outliers. *COIL2000*³ is a dataset for categorizing customers of an insurance company from dimensions of average age, customer main type, contribution private third party insurance, and number of mobile home policies. For multidimensional anomaly detection, we define those with an average age between 20 and 30 years old, the customer type as an hedonist, no private third party insurance contributions, and the number of mobile home policies as zero as normal samples, and the rest as outliers. Similar to *COIL2000*, *TIC2000*⁴ is also a dataset for categorizing customers of an insurance company from dimensions of average age, contribution private third party insurance, and number of mobile home policies. Therefore, it is set in the same way as *COIL2000*. *Pain* [54] is a dataset for predicting the intensity of ten facial action units based on the geometric facial features of patients. These ten facial action units are considered as ten dimensions, and observer ratings were performed on a six-point Likert-type scale that ranged from no pain to strong pain. We define patients with no pain as normal samples and patients with pain as outliers. The description of real-world datasets is shown in Table I.

2) *Baselines*: We compare ALOE with shallow and deep anomaly detection methods.

²<https://archive.ics.uci.edu/ml/machine-learning-databases/solar-flare/flare.data1>

³<https://archive.ics.uci.edu/dataset/125/insurance+company+benchmark+coil+2000>

⁴<https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients>

TABLE II
HS (MEAN $\pm$ STD.) OF EACH COMPARED METHOD ON THE BENCHMARK DATASETS

	Edm	Jura	Enb	Flare1	WQanimals	WQplants	Voice	Song	BeLaE	COIL2000	TIC2000	Pain
DIF	0.419 $\pm$ 0.032	0.368 $\pm$ 0.021	0.367 $\pm$ 0.075	0.118 $\pm$ 0.005	0.450 $\pm$ 0.009	0.427 $\pm$ 0.007	0.379 $\pm$ 0.018	0.348 $\pm$ 0.009	0.372 $\pm$ 0.007	0.371 $\pm$ 0.005	0.337 $\pm$ 0.000	0.090 $\pm$ 0.001
SLAD	0.435 $\pm$ 0.016	0.410 $\pm$ 0.006	0.474 $\pm$ 0.006	0.118 $\pm$ 0.036	0.437 $\pm$ 0.007	0.387 $\pm$ 0.005	0.635 $\pm$ 0.025	0.369 $\pm$ 0.002	0.465 $\pm$ 0.003	0.375 $\pm$ 0.001	0.355 $\pm$ 0.001	0.094 $\pm$ 0.001
ICL	0.403 $\pm$ 0.113	0.403 $\pm$ 0.028	0.508 $\pm$ 0.008	0.128 $\pm$ 0.005	0.441 $\pm$ 0.009	0.404 $\pm$ 0.011	0.617 $\pm$ 0.007	0.378 $\pm$ 0.005	0.428 $\pm$ 0.007	0.376 $\pm$ 0.000	0.355 $\pm$ 0.001	0.097 $\pm$ 0.002
RCA	0.403 $\pm$ 0.048	0.438 $\pm$ 0.007	0.412 $\pm$ 0.036	0.103 $\pm$ 0.041	0.458 $\pm$ 0.014	0.410 $\pm$ 0.002	0.398 $\pm$ 0.003	0.348 $\pm$ 0.014	0.394 $\pm$ 0.000	0.373 $\pm$ 0.002	0.338 $\pm$ 0.005	0.087 $\pm$ 0.001
RDP	0.306 $\pm$ 0.016	0.431 $\pm$ 0.014	0.425 $\pm$ 0.029	0.123 $\pm$ 0.051	0.452 $\pm$ 0.011	0.407 $\pm$ 0.009	0.487 $\pm$ 0.002	0.372 $\pm$ 0.016	0.418 $\pm$ 0.004	0.368 $\pm$ 0.008	0.353 $\pm$ 0.004	0.090 $\pm$ 0.010
GOAD	0.403 $\pm$ 0.016	0.465 $\pm$ 0.049	0.393 $\pm$ 0.023	0.138 $\pm$ 0.005	0.408 $\pm$ 0.007	0.371 $\pm$ 0.016	0.678 $\pm$ 0.006	0.401 $\pm$ 0.010	0.492 $\pm$ 0.003	0.397 $\pm$ 0.001	0.365 $\pm$ 0.001	0.099 $\pm$ 0.003
NeuTraL	0.468 $\pm$ 0.048	0.458 $\pm$ 0.028	0.474 $\pm$ 0.013	0.108 $\pm$ 0.015	0.437 $\pm$ 0.008	0.399 $\pm$ 0.019	0.607 $\pm$ 0.005	0.348 $\pm$ 0.009	0.496 $\pm$ 0.020	0.385 $\pm$ 0.017	0.353 $\pm$ 0.002	0.089 $\pm$ 0.013
REPEN	0.355 $\pm$ 0.065	0.431 $\pm$ 0.028	0.351 $\pm$ 0.026	0.128 $\pm$ 0.026	0.464 $\pm$ 0.005	0.417 $\pm$ 0.006	0.454 $\pm$ 0.021	0.395 $\pm$ 0.001	0.446 $\pm$ 0.012	0.374 $\pm$ 0.001	0.342 $\pm$ 0.000	0.091 $\pm$ 0.001
ALAD	0.355 $\pm$ 0.016	0.382 $\pm$ 0.007	0.406 $\pm$ 0.003	0.118 $\pm$ 0.010	0.458 $\pm$ 0.006	0.439 $\pm$ 0.007	0.327 $\pm$ 0.006	0.361 $\pm$ 0.002	0.377 $\pm$ 0.001	0.369 $\pm$ 0.002	0.337 $\pm$ 0.004	0.085 $\pm$ 0.003
AutoEncoder	0.339 $\pm$ 0.097	0.486 $\pm$ 0.021	0.477 $\pm$ 0.049	0.103 $\pm$ 0.005	0.463 $\pm$ 0.009	0.395 $\pm$ 0.016	0.330 $\pm$ 0.015	0.369 $\pm$ 0.002	0.431 $\pm$ 0.007	0.398 $\pm$ 0.002	0.371 $\pm$ 0.001	0.094 $\pm$ 0.001
CBLOF	0.387 $\pm$ 0.000	0.403 $\pm$ 0.000	0.416 $\pm$ 0.006	0.092 $\pm$ 0.015	0.462 $\pm$ 0.007	0.419 $\pm$ 0.016	0.348 $\pm$ 0.003	0.352 $\pm$ 0.004	0.385 $\pm$ 0.001	0.386 $\pm$ 0.001	0.362 $\pm$ 0.002	0.092 $\pm$ 0.007
COF	0.371 $\pm$ 0.048	0.444 $\pm$ 0.042	0.364 $\pm$ 0.146	0.095 $\pm$ 0.005	0.458 $\pm$ 0.006	0.399 $\pm$ 0.018	0.488 $\pm$ 0.013	0.369 $\pm$ 0.007	0.424 $\pm$ 0.005	0.363 $\pm$ 0.002	0.329 $\pm$ 0.002	0.093 $\pm$ 0.000
COPOD	0.363 $\pm$ 0.032	0.375 $\pm$ 0.042	0.476 $\pm$ 0.026	0.082 $\pm$ 0.010	0.443 $\pm$ 0.018	0.400 $\pm$ 0.008	0.514 $\pm$ 0.002	0.369 $\pm$ 0.006	0.418 $\pm$ 0.001	0.373 $\pm$ 0.002	0.342 $\pm$ 0.003	0.082 $\pm$ 0.002
DeepSVDD	0.371 $\pm$ 0.016	0.382 $\pm$ 0.007	0.393 $\pm$ 0.013	0.105 $\pm$ 0.015	0.441 $\pm$ 0.012	0.388 $\pm$ 0.003	0.354 $\pm$ 0.010	0.342 $\pm$ 0.001	0.423 $\pm$ 0.001	0.369 $\pm$ 0.009	0.338 $\pm$ 0.003	0.092 $\pm$ 0.010
ECOD	0.452 $\pm$ 0.048	0.472 $\pm$ 0.042	0.416 $\pm$ 0.006	0.095 $\pm$ 0.018	0.446 $\pm$ 0.009	0.403 $\pm$ 0.001	0.520 $\pm$ 0.010	0.367 $\pm$ 0.004	0.458 $\pm$ 0.002	0.375 $\pm$ 0.002	0.347 $\pm$ 0.002	0.086 $\pm$ 0.000
FB	0.282 $\pm$ 0.024	0.396 $\pm$ 0.014	0.481 $\pm$ 0.037	0.118 $\pm$ 0.010	0.468 $\pm$ 0.013	0.444 $\pm$ 0.001	0.348 $\pm$ 0.021	0.363 $\pm$ 0.008	0.401 $\pm$ 0.000	0.363 $\pm$ 0.007	0.341 $\pm$ 0.005	0.090 $\pm$ 0.004
GMM	0.339 $\pm$ 0.016	0.361 $\pm$ 0.021	0.458 $\pm$ 0.016	0.128 $\pm$ 0.003	0.456 $\pm$ 0.011	0.408 $\pm$ 0.008	0.488 $\pm$ 0.021	0.386 $\pm$ 0.001	0.397 $\pm$ 0.002	0.386 $\pm$ 0.001	0.344 $\pm$ 0.003	0.095 $\pm$ 0.002
HBOS	0.452 $\pm$ 0.032	0.382 $\pm$ 0.028	0.367 $\pm$ 0.091	0.128 $\pm$ 0.013	0.444 $\pm$ 0.019	0.406 $\pm$ 0.000	0.443 $\pm$ 0.007	0.399 $\pm$ 0.004	0.426 $\pm$ 0.015	0.362 $\pm$ 0.008	0.338 $\pm$ 0.001	0.095 $\pm$ 0.010
IForest	0.452 $\pm$ 0.073	0.451 $\pm$ 0.021	0.312 $\pm$ 0.019	0.054 $\pm$ 0.005	0.451 $\pm$ 0.002	0.388 $\pm$ 0.001	0.375 $\pm$ 0.001	0.361 $\pm$ 0.002	0.436 $\pm$ 0.003	0.382 $\pm$ 0.000	0.347 $\pm$ 0.002	0.094 $\pm$ 0.002
INNE	0.468 $\pm$ 0.048	0.375 $\pm$ 0.042	0.386 $\pm$ 0.049	0.085 $\pm$ 0.005	0.461 $\pm$ 0.010	0.408 $\pm$ 0.006	0.385 $\pm$ 0.004	0.346 $\pm$ 0.008	0.397 $\pm$ 0.003	0.360 $\pm$ 0.005	0.339 $\pm$ 0.004	0.088 $\pm$ 0.005
KDE	0.387 $\pm$ 0.032	0.354 $\pm$ 0.007	0.461 $\pm$ 0.010	0.133 $\pm$ 0.036	0.433 $\pm$ 0.007	0.405 $\pm$ 0.001	0.394 $\pm$ 0.011	0.340 $\pm$ 0.007	0.395 $\pm$ 0.006	0.359 $\pm$ 0.001	0.341 $\pm$ 0.004	0.096 $\pm$ 0.002
KNN	0.403 $\pm$ 0.032	0.354 $\pm$ 0.035	0.351 $\pm$ 0.026	0.108 $\pm$ 0.008	0.435 $\pm$ 0.001	0.392 $\pm$ 0.011	0.463 $\pm$ 0.004	0.376 $\pm$ 0.017	0.419 $\pm$ 0.005	0.366 $\pm$ 0.002	0.343 $\pm$ 0.002	0.089 $\pm$ 0.001
KPCA	0.435 $\pm$ 0.016	0.438 $\pm$ 0.014	0.253 $\pm$ 0.029	0.123 $\pm$ 0.003	0.449 $\pm$ 0.001	0.387 $\pm$ 0.001	0.494 $\pm$ 0.000	0.369 $\pm$ 0.017	0.401 $\pm$ 0.003	0.361 $\pm$ 0.003	0.328 $\pm$ 0.003	0.086 $\pm$ 0.003
LMDD	0.403 $\pm$ 0.065	0.389 $\pm$ 0.007	0.266 $\pm$ 0.019	0.128 $\pm$ 0.005	0.447 $\pm$ 0.003	0.391 $\pm$ 0.003	0.448 $\pm$ 0.007	0.376 $\pm$ 0.007	0.391 $\pm$ 0.007	0.394 $\pm$ 0.003	0.376 $\pm$ 0.002	0.086 $\pm$ 0.003
LODA	0.306 $\pm$ 0.048	0.424 $\pm$ 0.007	0.510 $\pm$ 0.039	0.113 $\pm$ 0.026	0.428 $\pm$ 0.005	0.381 $\pm$ 0.007	0.381 $\pm$ 0.011	0.397 $\pm$ 0.006	0.391 $\pm$ 0.002	0.381 $\pm$ 0.004	0.358 $\pm$ 0.001	0.090 $\pm$ 0.003
LOF	0.387 $\pm$ 0.048	0.465 $\pm$ 0.063	0.341 $\pm$ 0.026	0.118 $\pm$ 0.010	0.454 $\pm$ 0.007	0.431 $\pm$ 0.006	0.345 $\pm$ 0.007	0.378 $\pm$ 0.002	0.417 $\pm$ 0.006	0.383 $\pm$ 0.001	0.357 $\pm$ 0.001	0.090 $\pm$ 0.002
LSCP	0.403 $\pm$ 0.081	0.326 $\pm$ 0.014	0.416 $\pm$ 0.019	0.103 $\pm$ 0.036	0.443 $\pm$ 0.006	0.414 $\pm$ 0.009	0.490 $\pm$ 0.004	0.380 $\pm$ 0.003	0.395 $\pm$ 0.008	0.405 $\pm$ 0.001	0.329 $\pm$ 0.006	0.093 $\pm$ 0.004
LUNAR	0.387 $\pm$ 0.016	0.375 $\pm$ 0.021	0.396 $\pm$ 0.023	0.126 $\pm$ 0.041	0.452 $\pm$ 0.011	0.404 $\pm$ 0.015	0.496 $\pm$ 0.021	0.367 $\pm$ 0.008	0.427 $\pm$ 0.005	0.389 $\pm$ 0.001	0.367 $\pm$ 0.003	0.089 $\pm$ 0.004
MCD	0.444 $\pm$ 0.032	0.424 $\pm$ 0.007	0.269 $\pm$ 0.036	0.128 $\pm$ 0.021	0.429 $\pm$ 0.006	0.386 $\pm$ 0.020	0.501 $\pm$ 0.012	0.380 $\pm$ 0.010	0.412 $\pm$ 0.009	0.379 $\pm$ 0.002	0.348 $\pm$ 0.003	0.092 $\pm$ 0.001
OCSVM	0.371 $\pm$ 0.016	0.403 $\pm$ 0.021	0.370 $\pm$ 0.000	0.103 $\pm$ 0.010	0.454 $\pm$ 0.009	0.413 $\pm$ 0.007	0.365 $\pm$ 0.000	0.325 $\pm$ 0.010	0.405 $\pm$ 0.001	0.372 $\pm$ 0.005	0.344 $\pm$ 0.002	0.092 $\pm$ 0.007
PCA	0.427 $\pm$ 0.065	0.396 $\pm$ 0.014	0.208 $\pm$ 0.013	0.103 $\pm$ 0.008	0.442 $\pm$ 0.002	0.381 $\pm$ 0.001	0.463 $\pm$ 0.002	0.401 $\pm$ 0.006	0.434 $\pm$ 0.004	0.377 $\pm$ 0.002	0.349 $\pm$ 0.007	0.088 $\pm$ 0.001
Sampling	0.371 $\pm$ 0.089	0.389 $\pm$ 0.021	0.448 $\pm$ 0.029	0.118 $\pm$ 0.010	0.450 $\pm$ 0.018	0.418 $\pm$ 0.006	0.329 $\pm$ 0.015	0.340 $\pm$ 0.006	0.403 $\pm$ 0.014	0.363 $\pm$ 0.003	0.336 $\pm$ 0.002	0.098 $\pm$ 0.001
SOD	0.387 $\pm$ 0.081	0.444 $\pm$ 0.014	0.432 $\pm$ 0.026	0.133 $\pm$ 0.010	0.451 $\pm$ 0.024	0.397 $\pm$ 0.005	0.346 $\pm$ 0.004	0.352 $\pm$ 0.012	0.375 $\pm$ 0.002	0.358 $\pm$ 0.005	0.338 $\pm$ 0.001	0.091 $\pm$ 0.005
SUOD	0.379 $\pm$ 0.040	0.375 $\pm$ 0.021	0.416 $\pm$ 0.006	0.149 $\pm$ 0.021	0.454 $\pm$ 0.008	0.384 $\pm$ 0.001	0.463 $\pm$ 0.006	0.350 $\pm$ 0.010	0.453 $\pm$ 0.005	0.372 $\pm$ 0.005	0.342 $\pm$ 0.002	0.086 $\pm$ 0.005
ALOE	0.622$\pm$0.045	0.654$\pm$0.009	0.723$\pm$0.030	0.931$\pm$0.013	0.655$\pm$0.014	0.676$\pm$0.013	0.708$\pm$0.008	0.719$\pm$0.021	0.645$\pm$0.009	0.667$\pm$0.025	0.678$\pm$0.022	0.941$\pm$0.002

1) *Shallow*: CBLOF [55], COF [56], COPOD [57], ECOD [58], FB [59], GMM [5], HBOS [60], IForest [61], INNE [62], KDE [63], KNN [64], KPCA [63], LMDD [65], LODA [66], LOF [67], LSCP [68], MCD [69], OCSVM [70], PCA [60], Sampling [71], SOD [72], and SUOD [73].

2) *Deep*: DIF [74], SLAD [75], ICL [76], RCA [77], RDP [78], GOAD [79], NeuTraL [80], REPEN [81], ALAD [82], VAE [83], DeepSVDD [20], and LUNAR [84].

3) *Parameter Settings*: For ALOE, we use radial basis function with the same parameter settings as libsvm, and set $\lambda_1 = 0.2$, $\lambda_2 = 0.2$. For baselines, the parameters of all methods are set as in literature corresponding to the methods. Although these methods are not directly applicable to multidimensional anomaly detection, we can use them to identify outliers in testsets from each dimension and then evaluate the experimental results.

4) *Evaluation Metrics*: Since in multidimensional anomaly detection, each sample corresponds to a state vector, we need to use different evaluation metrics including Hamming Score (HS), Exact Match (EM), and Sub-EM (SEM). These evaluation metrics are widely used to assess the performance of models targeting multioutput tasks. Specifically, for HS, we have $HS = (1/p) \sum_{i=1}^p (1/m) \sum_{j=1}^m \mathbf{1}[y_{ij} = \hat{y}_{ij}]$, where $\mathbf{1}[a]$ returns 1 if a holds and 0 otherwise. For EM, we have $EM = (1/p) \sum_{i=1}^p \mathbf{1}[\sum_{j=1}^m \mathbf{1}[y_{ij} = \hat{y}_{ij}] == m]$. For SEM, we have $SEM = (1/p) \sum_{i=1}^p \mathbf{1}[\sum_{j=1}^m \mathbf{1}[y_{ij} = \hat{y}_{ij}] \geq m - 1]$. To more fully evaluate the performance of ALOE, we further calculate the average AUC among all dimensions.

B. Experimental Results

1) *Effectiveness Analysis*: HS is to quantify the average discrepancy between predicted labels and ground truth labels.

It essentially computes the proportion of errors occurring across all prediction-label pairs. However, this computation is conducted individually for each output, and the error proportions are then averaged. As presented in Table II, ALOE achieves the highest HS in all datasets, which demonstrates its superiority. Notably, on “Edm,” “Jura,” “Enb,” “Voice,” and “BeLaE” datasets, our method significantly outperforms other anomaly detection methods by 20%. In addition, on “Flare1” and “Pain” datasets, ALOE obtains the 0.9 HS while other baselines only have around 0.1 HS. These results indicate that our method exhibits a low level of inconsistency between its detections and the actual detection results for test samples under each dimension, thereby accurately identifying outliers from most dimensions. This good effectiveness can be attributed to the method’s capability to consider each dimension individually while further accounting for the interdependencies between these dimensions. This consideration ensures that the method is subject to tighter constraints during the training phase, resulting in a more precise optimal boundary learned for each dimension.

EM evaluates the precise correspondence between the detection results of an anomaly detection method and the ground truth labels for test samples across various dimensions. The method achieves EM if its multiple output results perfectly align with the ground truths. Table III presents EM of ALOE and the baselines across 12 real-world datasets, respectively. Experimental results indicate that our method achieves 0.845 EM on “Flare1” dataset, demonstrating its robust performance under the stringent accuracy requirement. Moreover, our method exhibits a 10%–15% improvement in EM compared to other baselines on “Jura,” “Enb,” “Voice,” and “Pain” datasets. This signifies that our method identifies a higher proportion of true anomalies among the test samples

TABLE III
EM (MEAN $\pm$ STD.) OF EACH COMPARED METHOD ON THE BENCHMARK DATASETS

	Edm	Jura	Enb	FlareI	WQanimals	WQplants	Voice	Song	BeLaE	COIL2000	TIC2000	Pain
DIF	0.290 $\pm$ 0.032	0.340 $\pm$ 0.021	0.406 $\pm$ 0.101	0.731 $\pm$ 0.014	0.028 $\pm$ 0.021	0.068 $\pm$ 0.025	0.382 $\pm$ 0.013	0.261 $\pm$ 0.034	0.171 $\pm$ 0.000	0.084 $\pm$ 0.013	0.282 $\pm$ 0.003	0.657 $\pm$ 0.008
SLAD	0.274 $\pm$ 0.048	0.340 $\pm$ 0.035	0.350 $\pm$ 0.010	0.738 $\pm$ 0.002	0.050 $\pm$ 0.012	0.085 $\pm$ 0.000	0.147 $\pm$ 0.014	0.268 $\pm$ 0.018	0.102 $\pm$ 0.000	0.087 $\pm$ 0.015	0.249 $\pm$ 0.004	0.638 $\pm$ 0.001
ICL	0.339 $\pm$ 0.145	0.340 $\pm$ 0.035	0.247 $\pm$ 0.006	0.715 $\pm$ 0.005	0.024 $\pm$ 0.006	0.052 $\pm$ 0.029	0.148 $\pm$ 0.017	0.245 $\pm$ 0.001	0.126 $\pm$ 0.004	0.086 $\pm$ 0.006	0.252 $\pm$ 0.001	0.608 $\pm$ 0.006
RCA	0.306 $\pm$ 0.048	0.236 $\pm$ 0.014	0.364 $\pm$ 0.052	0.792 $\pm$ 0.005	0.038 $\pm$ 0.012	0.047 $\pm$ 0.022	0.386 $\pm$ 0.000	0.271 $\pm$ 0.005	0.163 $\pm$ 0.001	0.083 $\pm$ 0.002	0.276 $\pm$ 0.005	0.672 $\pm$ 0.001
RDP	0.403 $\pm$ 0.016	0.292 $\pm$ 0.000	0.305 $\pm$ 0.000	0.762 $\pm$ 0.012	0.050 $\pm$ 0.002	0.087 $\pm$ 0.045	0.293 $\pm$ 0.013	0.236 $\pm$ 0.006	0.101 $\pm$ 0.003	0.096 $\pm$ 0.004	0.254 $\pm$ 0.001	0.620 $\pm$ 0.003
GOAD	0.290 $\pm$ 0.032	0.229 $\pm$ 0.049	0.351 $\pm$ 0.019	0.692 $\pm$ 0.002	0.045 $\pm$ 0.006	0.101 $\pm$ 0.003	0.107 $\pm$ 0.004	0.220 $\pm$ 0.013	0.087 $\pm$ 0.001	0.069 $\pm$ 0.027	0.233 $\pm$ 0.001	0.633 $\pm$ 0.010
NeuTraL	0.226 $\pm$ 0.032	0.243 $\pm$ 0.049	0.250 $\pm$ 0.029	0.777 $\pm$ 0.002	0.054 $\pm$ 0.005	0.071 $\pm$ 0.038	0.183 $\pm$ 0.008	0.299 $\pm$ 0.004	0.073 $\pm$ 0.012	0.085 $\pm$ 0.008	0.257 $\pm$ 0.003	0.590 $\pm$ 0.005
REPEN	0.403 $\pm$ 0.048	0.264 $\pm$ 0.042	0.409 $\pm$ 0.026	0.738 $\pm$ 0.005	0.028 $\pm$ 0.025	0.026 $\pm$ 0.038	0.312 $\pm$ 0.005	0.255 $\pm$ 0.006	0.080 $\pm$ 0.000	0.088 $\pm$ 0.002	0.270 $\pm$ 0.001	0.649 $\pm$ 0.002
ALAD	0.387 $\pm$ 0.048	0.313 $\pm$ 0.014	0.364 $\pm$ 0.006	0.785 $\pm$ 0.021	0.024 $\pm$ 0.010	0.042 $\pm$ 0.057	0.463 $\pm$ 0.006	0.290 $\pm$ 0.017	0.150 $\pm$ 0.010	0.080 $\pm$ 0.003	0.280 $\pm$ 0.009	0.664 $\pm$ 0.003
AutoEncoder	0.419 $\pm$ 0.097	0.278 $\pm$ 0.007	0.282 $\pm$ 0.078	0.738 $\pm$ 0.005	0.021 $\pm$ 0.035	0.040 $\pm$ 0.016	0.449 $\pm$ 0.010	0.229 $\pm$ 0.004	0.067 $\pm$ 0.006	0.069 $\pm$ 0.006	0.213 $\pm$ 0.005	0.489 $\pm$ 0.011
CBLOF	0.355 $\pm$ 0.000	0.285 $\pm$ 0.014	0.341 $\pm$ 0.029	0.815 $\pm$ 0.002	0.045 $\pm$ 0.011	0.038 $\pm$ 0.022	0.420 $\pm$ 0.012	0.290 $\pm$ 0.017	0.162 $\pm$ 0.002	0.079 $\pm$ 0.000	0.249 $\pm$ 0.001	0.654 $\pm$ 0.000
COF	0.387 $\pm$ 0.065	0.264 $\pm$ 0.069	0.380 $\pm$ 0.159	0.792 $\pm$ 0.002	0.033 $\pm$ 0.021	0.080 $\pm$ 0.000	0.295 $\pm$ 0.010	0.268 $\pm$ 0.022	0.139 $\pm$ 0.004	0.085 $\pm$ 0.002	0.298$\pm$0.004	0.658 $\pm$ 0.009
COPOD	0.419 $\pm$ 0.097	0.340 $\pm$ 0.076	0.276 $\pm$ 0.003	0.808 $\pm$ 0.002	0.040 $\pm$ 0.006	0.083 $\pm$ 0.016	0.257 $\pm$ 0.009	0.255 $\pm$ 0.032	0.133 $\pm$ 0.001	0.078 $\pm$ 0.009	0.268 $\pm$ 0.001	0.688 $\pm$ 0.001
DeepSVDD	0.387 $\pm$ 0.048	0.326 $\pm$ 0.021	0.344 $\pm$ 0.010	0.777 $\pm$ 0.007	0.047 $\pm$ 0.006	0.083 $\pm$ 0.010	0.429 $\pm$ 0.006	0.283 $\pm$ 0.004	0.126 $\pm$ 0.005	0.091 $\pm$ 0.013	0.281 $\pm$ 0.002	0.633 $\pm$ 0.008
ECOD	0.274 $\pm$ 0.032	0.257 $\pm$ 0.056	0.373 $\pm$ 0.010	0.754 $\pm$ 0.000	0.028 $\pm$ 0.006	0.057 $\pm$ 0.025	0.252 $\pm$ 0.001	0.245 $\pm$ 0.002	0.052 $\pm$ 0.005	0.089 $\pm$ 0.001	0.271 $\pm$ 0.004	0.684 $\pm$ 0.002
FB	0.516$\pm$0.065	0.333 $\pm$ 0.035	0.302 $\pm$ 0.049	0.731 $\pm$ 0.007	0.024 $\pm$ 0.023	0.035 $\pm$ 0.051	0.420 $\pm$ 0.004	0.287 $\pm$ 0.014	0.140 $\pm$ 0.004	0.088 $\pm$ 0.007	0.273 $\pm$ 0.001	0.660 $\pm$ 0.002
GMM	0.403 $\pm$ 0.000	0.424 $\pm$ 0.035	0.266 $\pm$ 0.032	0.708 $\pm$ 0.028	0.031 $\pm$ 0.034	0.061 $\pm$ 0.016	0.287 $\pm$ 0.019	0.252 $\pm$ 0.010	0.137 $\pm$ 0.002	0.081 $\pm$ 0.011	0.263 $\pm$ 0.001	0.530 $\pm$ 0.002
HBOS	0.290 $\pm$ 0.048	0.361 $\pm$ 0.021	0.412 $\pm$ 0.114	0.738 $\pm$ 0.005	0.061 $\pm$ 0.018	0.080 $\pm$ 0.003	0.347 $\pm$ 0.005	0.236 $\pm$ 0.018	0.127 $\pm$ 0.003	0.086 $\pm$ 0.018	0.281 $\pm$ 0.007	0.654 $\pm$ 0.001
IFloss	0.306 $\pm$ 0.129	0.271 $\pm$ 0.014	0.442 $\pm$ 0.010	0.845$\pm$0.014	0.038 $\pm$ 0.001	0.073 $\pm$ 0.000	0.396 $\pm$ 0.003	0.280 $\pm$ 0.010	0.117 $\pm$ 0.001	0.084 $\pm$ 0.011	0.268 $\pm$ 0.001	0.661 $\pm$ 0.009
INNE	0.210 $\pm$ 0.081	0.319 $\pm$ 0.049	0.377 $\pm$ 0.039	0.800 $\pm$ 0.017	0.050 $\pm$ 0.005	0.042 $\pm$ 0.035	0.389 $\pm$ 0.018	0.312 $\pm$ 0.004	0.152 $\pm$ 0.003	0.096 $\pm$ 0.008	0.287 $\pm$ 0.001	0.661 $\pm$ 0.006
KDE	0.290 $\pm$ 0.000	0.382 $\pm$ 0.028	0.312 $\pm$ 0.003	0.738 $\pm$ 0.009	0.045 $\pm$ 0.006	0.094 $\pm$ 0.038	0.396 $\pm$ 0.018	0.322$\pm$0.013	0.152 $\pm$ 0.002	0.093 $\pm$ 0.009	0.273 $\pm$ 0.001	0.651 $\pm$ 0.001
KNN	0.306 $\pm$ 0.048	0.403 $\pm$ 0.035	0.396 $\pm$ 0.019	0.808 $\pm$ 0.014	0.050 $\pm$ 0.013	0.052 $\pm$ 0.013	0.327 $\pm$ 0.016	0.252 $\pm$ 0.003	0.133 $\pm$ 0.003	0.093 $\pm$ 0.001	0.267 $\pm$ 0.000	0.655 $\pm$ 0.005
KPCA	0.258 $\pm$ 0.048	0.292 $\pm$ 0.028	0.536 $\pm$ 0.026	0.754 $\pm$ 0.002	0.054 $\pm$ 0.004	0.073 $\pm$ 0.000	0.279 $\pm$ 0.017	0.274 $\pm$ 0.001	0.142 $\pm$ 0.003	0.091 $\pm$ 0.009	0.290 $\pm$ 0.004	0.667 $\pm$ 0.010
LMDD	0.306 $\pm$ 0.097	0.326 $\pm$ 0.021	0.510 $\pm$ 0.045	0.754 $\pm$ 0.017	0.054 $\pm$ 0.006	0.104$\pm$0.010	0.335 $\pm$ 0.009	0.255 $\pm$ 0.013	0.167 $\pm$ 0.004	0.072 $\pm$ 0.021	0.228 $\pm$ 0.002	0.683 $\pm$ 0.002
LODA	0.500 $\pm$ 0.081	0.278 $\pm$ 0.014	0.247 $\pm$ 0.058	0.777 $\pm$ 0.012	0.047 $\pm$ 0.017	0.085 $\pm$ 0.003	0.398 $\pm$ 0.009	0.220 $\pm$ 0.013	0.159 $\pm$ 0.006	0.090 $\pm$ 0.003	0.261 $\pm$ 0.005	0.666 $\pm$ 0.004
LOF	0.323 $\pm$ 0.065	0.229 $\pm$ 0.083	0.406 $\pm$ 0.049	0.700 $\pm$ 0.026	0.019 $\pm$ 0.004	0.031 $\pm$ 0.013	0.443 $\pm$ 0.000	0.248 $\pm$ 0.030	0.118 $\pm$ 0.004	0.086 $\pm$ 0.001	0.246 $\pm$ 0.012	0.669 $\pm$ 0.008
LSCP	0.306 $\pm$ 0.113	0.424 $\pm$ 0.021	0.347 $\pm$ 0.042	0.762 $\pm$ 0.012	0.061$\pm$0.017	0.061 $\pm$ 0.025	0.297 $\pm$ 0.004	0.248 $\pm$ 0.013	0.152 $\pm$ 0.000	0.073 $\pm$ 0.002	0.297 $\pm$ 0.005	0.649 $\pm$ 0.005
LUNAR	0.306 $\pm$ 0.016	0.347 $\pm$ 0.000	0.380 $\pm$ 0.006	0.746 $\pm$ 0.014	0.054 $\pm$ 0.022	0.059 $\pm$ 0.016	0.278 $\pm$ 0.001	0.261 $\pm$ 0.013	0.135 $\pm$ 0.001	0.073 $\pm$ 0.020	0.276 $\pm$ 0.004	0.646 $\pm$ 0.004
MCD	0.258 $\pm$ 0.000	0.313 $\pm$ 0.021	0.516 $\pm$ 0.036	0.731 $\pm$ 0.000	0.054 $\pm$ 0.032	0.078 $\pm$ 0.045	0.275 $\pm$ 0.012	0.232 $\pm$ 0.020	0.146 $\pm$ 0.003	0.091 $\pm$ 0.006	0.273 $\pm$ 0.001	0.659 $\pm$ 0.003
OCSVM	0.371 $\pm$ 0.048	0.326 $\pm$ 0.035	0.370 $\pm$ 0.010	0.777 $\pm$ 0.005	0.040 $\pm$ 0.004	0.042 $\pm$ 0.013	0.418 $\pm$ 0.006	0.312 $\pm$ 0.013	0.133 $\pm$ 0.006	0.093 $\pm$ 0.006	0.276 $\pm$ 0.002	0.659 $\pm$ 0.003
PCA	0.290 $\pm$ 0.113	0.326 $\pm$ 0.000	0.520 $\pm$ 0.029	0.769 $\pm$ 0.002	0.047 $\pm$ 0.002	0.092 $\pm$ 0.006	0.325 $\pm$ 0.005	0.239 $\pm$ 0.000	0.113 $\pm$ 0.002	0.088 $\pm$ 0.008	0.262 $\pm$ 0.006	0.657 $\pm$ 0.001
Sampling	0.323 $\pm$ 0.097	0.319 $\pm$ 0.035	0.305 $\pm$ 0.042	0.746 $\pm$ 0.012	0.052 $\pm$ 0.014	0.054 $\pm$ 0.022	0.455 $\pm$ 0.001	0.293 $\pm$ 0.013	0.145 $\pm$ 0.001	0.090 $\pm$ 0.013	0.279 $\pm$ 0.004	0.645 $\pm$ 0.001
SOD	0.323 $\pm$ 0.081	0.278 $\pm$ 0.049	0.328 $\pm$ 0.026	0.723 $\pm$ 0.007	0.033 $\pm$ 0.010	0.066 $\pm$ 0.013	0.428 $\pm$ 0.017	0.268 $\pm$ 0.013	0.174 $\pm$ 0.003	0.098 $\pm$ 0.013	0.277 $\pm$ 0.009	0.666 $\pm$ 0.010
SUOD	0.339 $\pm$ 0.097	0.368 $\pm$ 0.035	0.321 $\pm$ 0.010	0.708 $\pm$ 0.024	0.038 $\pm$ 0.018	0.090 $\pm$ 0.010	0.316 $\pm$ 0.004	0.318 $\pm$ 0.020	0.101 $\pm$ 0.002	0.084 $\pm$ 0.020	0.276 $\pm$ 0.010	0.674 $\pm$ 0.006
ALOE	0.327 $\pm$ 0.067	0.468$\pm$0.021	0.547$\pm$0.038	0.845$\pm$0.032	0.050 $\pm$ 0.023	0.062 $\pm$ 0.016	0.474$\pm$0.042	0.304 $\pm$ 0.012	0.177$\pm$0.030	0.099$\pm$0.006	0.285 $\pm$ 0.010	0.734$\pm$0.009

TABLE IV
SEM (MEAN $\pm$ STD.) OF EACH COMPARED METHOD ON THE BENCHMARK DATASETS

	Edm	Jura	Enb	FlareI	WQanimals	WQplants	Voice	Song	BeLaE	COIL2000	TIC2000	Pain
DIF	0.571±0.032	0.624±0.021	0.560±0.049	0.692±0.008	0.681±0.005	0.681±0.005	0.560±0.014	0.659±0.003	0.682±0.003	0.697±0.004	0.676±0.000	0.695±0.002
SLAD	0.555±0.016	0.540±0.007	0.502±0.023	0.685±0.015	0.686±0.000	0.693±0.002	0.282±0.039	0.636±0.006	0.643±0.005	0.696±0.000	0.669±0.000	0.696±0.005
ICL	0.555±0.081	0.554±0.021	0.437±0.023	0.685±0.000	0.695±0.000	0.695±0.005	0.318±0.008	0.662±0.006	0.672±0.008	0.695±0.001	0.673±0.001	0.699±0.004
RCA	0.587±0.048	0.589±0.028	0.512±0.019	0.677±0.023	0.679±0.007	0.693±0.002	0.518±0.006	0.633±0.003	0.660±0.009	0.697±0.003	0.676±0.001	0.694±0.002
RDP	0.684±0.016	0.547±0.028	0.544±0.058	0.685±0.015	0.676±0.000	0.681±0.000	0.432±0.005	0.652±0.003	0.678±0.001	0.697±0.000	0.670±0.001	0.697±0.004
GOAD	0.603±0.000	0.540±0.049	0.564±0.026	0.685±0.015	0.691±0.009	0.698±0.002	0.236±0.007	0.649±0.006	0.612±0.010	0.695±0.002	0.675±0.003	0.694±0.001
NeuTraL	0.539±0.065	0.540±0.007	0.502±0.003	0.700±0.000	0.688±0.002	0.695±0.005	0.304±0.005	0.659±0.010	0.629±0.004	0.693±0.002	0.669±0.007	0.700±0.001
REPEN	0.587±0.081	0.575±0.014	0.590±0.026	0.685±0.000	0.688±0.002	0.698±0.002	0.480±0.016	0.617±0.019	0.675±0.004	0.695±0.001	0.671±0.000	0.692±0.001
ALAD	0.603±0.016	0.624±0.000	0.525±0.000	0.677±0.000	0.686±0.000	0.686±0.000	0.582±0.002	0.624±0.003	0.674±0.003	0.697±0.000	0.672±0.000	0.694±0.000
AutoEncoder	0.603±0.097	0.450±0.035	0.463±0.019	0.700±0.023	0.686±0.000	0.693±0.005	0.592±0.005	0.655±0.013	0.677±0.005	0.695±0.002	0.674±0.000	0.700±0.003
CBLOF	0.571±0.000	0.610±0.014	0.528±0.016	0.685±0.008	0.686±0.000	0.695±0.005	0.584±0.005	0.617±0.013	0.673±0.001	0.695±0.000	0.664±0.001	0.693±0.001
COF	0.571±0.032	0.547±0.014	0.593±0.133	0.692±0.000	0.688±0.009	0.688±0.002	0.429±0.005	0.614±0.013	0.642±0.003	0.697±0.007	0.671±0.001	0.690±0.000
COPOD	0.555±0.032	0.610±0.007	0.473±0.049	0.692±0.000	0.679±0.005	0.674±0.005	0.414±0.002	0.620±0.006	0.662±0.006	0.696±0.002	0.674±0.001	0.692±0.001
DeepSVDD	0.571±0.016	0.610±0.007	0.570±0.016	0.692±0.000	0.686±0.002	0.695±0.005	0.564±0.014	0.643±0.010	0.653±0.004	0.697±0.003	0.671±0.000	0.698±0.003
ECOD	0.523±0.065	0.499±0.028	0.495±0.003	0.692±0.008	0.688±0.002	0.686±0.002	0.408±0.015	0.655±0.016	0.674±0.006	0.696±0.002	0.672±0.001	0.696±0.004
FB	0.619±0.016	0.575±0.007	0.437±0.026	0.700±0.008	0.683±0.002	0.662±0.002	0.585±0.018	0.624±0.003	0.661±0.001	0.696±0.010	0.673±0.007	0.691±0.003
GMM	0.619±0.032	0.554±0.007	0.518±0.000	0.700±0.008	0.695±0.000	0.676±0.000	0.437±0.007	0.624±0.013	0.655±0.008	0.695±0.002	0.672±0.000	0.700±0.004
HBOS	0.506±0.016	0.575±0.035	0.554±0.068	0.685±0.008	0.688±0.002	0.683±0.000	0.468±0.004	0.639±0.013	0.647±0.003	0.697±0.000	0.673±0.001	0.690±0.001
IForest	0.490±0.016	0.526±0.028	0.635±0.029	0.700±0.000	0.688±0.012	0.693±0.009	0.554±0.002	0.617±0.013	0.640±0.005	0.695±0.001	0.667±0.001	0.694±0.003
INNE	0.555±0.016	0.631±0.035	0.551±0.058	0.692±0.000	0.688±0.000	0.691±0.014	0.540±0.003	0.643±0.013	0.669±0.004	0.696±0.007	0.669±0.000	0.694±0.002
KDE	0.635±0.065	0.610±0.014	0.466±0.016	0.692±0.015	0.695±0.002	0.693±0.002	0.516±0.016	0.633±0.010	0.669±0.004	0.697±0.002	0.672±0.000	0.690±0.000
KNN	0.587±0.016	0.589±0.035	0.603±0.032	0.669±0.000	0.683±0.002	0.695±0.002	0.446±0.021	0.630±0.006	0.634±0.008	0.695±0.002	0.673±0.000	0.694±0.003
KPCA	0.571±0.016	0.533±0.000	0.658±0.032	0.677±0.008	0.681±0.002	0.691±0.000	0.432±0.004	0.617±0.013	0.661±0.013	0.696±0.001	0.672±0.001	0.695±0.002
LMDD	0.587±0.032	0.596±0.007	0.658±0.006	0.692±0.008	0.683±0.005	0.686±0.002	0.468±0.020	0.633±0.003	0.661±0.010	0.695±0.001	0.670±0.000	0.694±0.003
LODA	0.587±0.016	0.575±0.028	0.434±0.019	0.685±0.015	0.693±0.007	0.691±0.005	0.540±0.006	0.630±0.013	0.664±0.004	0.695±0.003	0.670±0.000	0.693±0.001
LOF	0.603±0.032	0.540±0.042	0.612±0.003	0.700±0.023	0.683±0.009	0.698±0.009	0.568±0.010	0.630±0.006	0.675±0.005	0.696±0.003	0.673±0.001	0.694±0.002
LSCP	0.587±0.048	0.624±0.007	0.521±0.003	0.700±0.008	0.683±0.012	0.686±0.009	0.422±0.009	0.633±0.010	0.655±0.008	0.694±0.001	0.675±0.000	0.691±0.002
LUNAR	0.619±0.048	0.603±0.042	0.528±0.052	0.685±0.015	0.676±0.002	0.688±0.000	0.430±0.020	0.624±0.006	0.631±0.003	0.696±0.009	0.675±0.001	0.695±0.002
MCD	0.555±0.065	0.540±0.007	0.645±0.036	0.669±0.000	0.695±0.007	0.688±0.002	0.424±0.008	0.652±0.000	0.643±0.001	0.694±0.001	0.669±0.000	0.693±0.002
OCSVM	0.587±0.016	0.568±0.007	0.590±0.010	0.700±0.000	0.679±0.002	0.691±0.005	0.551±0.004	0.636±0.010	0.655±0.004	0.692±0.006	0.667±0.003	0.696±0.001
PCA	0.555±0.016	0.582±0.028	0.664±0.003	0.692±0.015	0.679±0.005	0.693±0.002	0.450±0.002	0.627±0.006	0.639±0.001	0.696±0.003	0.671±0.001	0.697±0.000
Sampling	0.635±0.081	0.603±0.007	0.499±0.016	0.692±0.031	0.686±0.000	0.686±0.002	0.588±0.017	0.643±0.003	0.672±0.008	0.694±0.001	0.673±0.002	0.690±0.003
SOD	0.603±0.081	0.533±0.021	0.508±0.026	0.700±0.000	0.695±0.007	0.693±0.005	0.581±0.002	0.646±0.013	0.666±0.009	0.696±0.003	0.677±0.000	0.692±0.009
SUOD	0.603±0.016	0.582±0.007	0.547±0.003	0.669±0.008	0.681±0.002	0.686±0.002	0.459±0.006	0.643±0.010	0.647±0.009	0.694±0.009	0.669±0.000	0.693±0.003
ALOE	0.918±0.033	0.841±0.008	0.900±0.007	0.950±0.007	0.247±0.005	0.294±0.010	0.943±0.004	0.855±0.005	0.448±0.012	0.439±0.008	0.768±0.007	0.823±0.002

TABLE V
AVERAGE AUC (MEAN $\pm$ STD.) OF EACH COMPARED METHOD ON THE BENCHMARK DATASETS

	Edm	Jura	Enb	Flare1	WQanimals	WQplants	Voice	Song	BeLaE	COIL2000	TIC2000	Pain
DIF	0.521 $\pm$ 0.011	0.632 $\pm$ 0.039	0.646 $\pm$ 0.056	0.472 $\pm$ 0.001	0.377 $\pm$ 0.003	0.505 $\pm$ 0.011	0.667 $\pm$ 0.021	0.548 $\pm$ 0.004	0.623$\pm$0.003	0.550 $\pm$ 0.001	0.540 $\pm$ 0.009	0.673 $\pm$ 0.005
SLAD	0.465 $\pm$ 0.033	0.558 $\pm$ 0.037	0.507 $\pm$ 0.003	0.000 $\pm$ 0.000	0.422 $\pm$ 0.002	0.586 $\pm$ 0.012	0.300 $\pm$ 0.028	0.509 $\pm$ 0.010	0.495 $\pm$ 0.000	0.542 $\pm$ 0.002	0.537 $\pm$ 0.001	0.671 $\pm$ 0.014
ICL	0.363 $\pm$ 0.040	0.546 $\pm$ 0.004	0.532 $\pm$ 0.019	0.613 $\pm$ 0.039	0.419 $\pm$ 0.013	0.534 $\pm$ 0.012	0.341 $\pm$ 0.000	0.480 $\pm$ 0.018	0.554 $\pm$ 0.004	0.533 $\pm$ 0.004	0.511 $\pm$ 0.010	0.463 $\pm$ 0.005
RCA	0.531 $\pm$ 0.002	0.525 $\pm$ 0.043	0.590 $\pm$ 0.049	0.668 $\pm$ 0.066	0.379 $\pm$ 0.003	0.540 $\pm$ 0.006	0.641 $\pm$ 0.005	0.566 $\pm$ 0.015	0.586 $\pm$ 0.022	0.548 $\pm$ 0.005	0.545 $\pm$ 0.005	0.704 $\pm$ 0.004
RDP	0.349 $\pm$ 0.082	0.551 $\pm$ 0.018	0.526 $\pm$ 0.021	0.595 $\pm$ 0.005	0.401 $\pm$ 0.011	0.539 $\pm$ 0.007	0.520 $\pm$ 0.011	0.519 $\pm$ 0.012	0.552 $\pm$ 0.024	0.552 $\pm$ 0.000	0.530 $\pm$ 0.003	0.671 $\pm$ 0.008
GOAD	0.443 $\pm$ 0.027	0.483 $\pm$ 0.028	0.608 $\pm$ 0.028	0.000 $\pm$ 0.000	0.449 $\pm$ 0.009	0.582 $\pm$ 0.004	0.228 $\pm$ 0.001	0.481 $\pm$ 0.014	0.443 $\pm$ 0.015	0.489 $\pm$ 0.003	0.495 $\pm$ 0.003	0.482 $\pm$ 0.000
NeuTraL	0.465 $\pm$ 0.010	0.544 $\pm$ 0.013	0.538 $\pm$ 0.009	0.000 $\pm$ 0.000	0.431 $\pm$ 0.000	0.574 $\pm$ 0.007	0.328 $\pm$ 0.018	0.521 $\pm$ 0.008	0.438 $\pm$ 0.012	0.523 $\pm$ 0.002	0.518 $\pm$ 0.023	0.593 $\pm$ 0.023
REPEN	0.559 $\pm$ 0.048	0.538 $\pm$ 0.036	0.669 $\pm$ 0.025	0.720 $\pm$ 0.128	0.383 $\pm$ 0.007	0.534 $\pm$ 0.001	0.572 $\pm$ 0.023	0.496 $\pm$ 0.014	0.532 $\pm$ 0.007	0.530 $\pm$ 0.008	0.532 $\pm$ 0.002	0.663 $\pm$ 0.004
ALAD	0.475 $\pm$ 0.024	0.597 $\pm$ 0.016	0.561 $\pm$ 0.000	0.623 $\pm$ 0.030	0.374 $\pm$ 0.007	0.497 $\pm$ 0.001	0.740 $\pm$ 0.003	0.546 $\pm$ 0.036	0.602 $\pm$ 0.006	0.552 $\pm$ 0.001	0.548 $\pm$ 0.000	0.679 $\pm$ 0.003
AutoEncoder	0.583 $\pm$ 0.001	0.488 $\pm$ 0.019	0.471 $\pm$ 0.039	0.000 $\pm$ 0.000	0.364 $\pm$ 0.004	0.548 $\pm$ 0.015	0.717 $\pm$ 0.015	0.551 $\pm$ 0.053	0.550 $\pm$ 0.004	0.509 $\pm$ 0.012	0.493 $\pm$ 0.012	0.576 $\pm$ 0.015
CBLOF	0.581 $\pm$ 0.043	0.612 $\pm$ 0.004	0.534 $\pm$ 0.015	0.707 $\pm$ 0.017	0.356 $\pm$ 0.009	0.498 $\pm$ 0.028	0.717 $\pm$ 0.010	0.539 $\pm$ 0.008	0.612 $\pm$ 0.007	0.519 $\pm$ 0.003	0.509 $\pm$ 0.000	0.685 $\pm$ 0.003
COF	0.602 $\pm$ 0.001	0.551 $\pm$ 0.024	0.651 $\pm$ 0.193	0.580 $\pm$ 0.036	0.388 $\pm$ 0.011	0.550 $\pm$ 0.037	0.526 $\pm$ 0.001	0.541 $\pm$ 0.049	0.545 $\pm$ 0.006	0.550 $\pm$ 0.011	0.539 $\pm$ 0.000	0.666 $\pm$ 0.004
COPOD	0.584 $\pm$ 0.008	0.601 $\pm$ 0.031	0.493 $\pm$ 0.034	0.000 $\pm$ 0.000	0.399 $\pm$ 0.005	0.544 $\pm$ 0.000	0.472 $\pm$ 0.009	0.542 $\pm$ 0.008	0.583 $\pm$ 0.015	0.554 $\pm$ 0.001	0.552 $\pm$ 0.002	0.724 $\pm$ 0.003
DeepSVDD	0.604 $\pm$ 0.033	0.588 $\pm$ 0.004	0.668 $\pm$ 0.004	0.000 $\pm$ 0.029	0.386 $\pm$ 0.015	0.559 $\pm$ 0.009	0.694 $\pm$ 0.006	0.555 $\pm$ 0.034	0.556 $\pm$ 0.008	0.545 $\pm$ 0.003	0.547 $\pm$ 0.005	0.682 $\pm$ 0.003
ECOD	0.488 $\pm$ 0.015	0.457 $\pm$ 0.003	0.591 $\pm$ 0.014	0.000 $\pm$ 0.080	0.407 $\pm$ 0.003	0.544 $\pm$ 0.004	0.464 $\pm$ 0.002	0.541 $\pm$ 0.010	0.489 $\pm$ 0.005	0.537 $\pm$ 0.004	0.526 $\pm$ 0.001	0.649 $\pm$ 0.009
FB	0.671$\pm$0.018	0.575 $\pm$ 0.023	0.495 $\pm$ 0.040	0.629 $\pm$ 0.126	0.355 $\pm$ 0.001	0.463 $\pm$ 0.007	0.718 $\pm$ 0.012	0.535 $\pm$ 0.005	0.604 $\pm$ 0.002	0.560 $\pm$ 0.005	0.542 $\pm$ 0.001	0.706 $\pm$ 0.004
GMM	0.586 $\pm$ 0.014	0.627 $\pm$ 0.012	0.528 $\pm$ 0.019	0.551 $\pm$ 0.133	0.387 $\pm$ 0.011	0.541 $\pm$ 0.016	0.502 $\pm$ 0.023	0.527 $\pm$ 0.023	0.583 $\pm$ 0.002	0.514 $\pm$ 0.002	0.509 $\pm$ 0.004	0.465 $\pm$ 0.013
HBOS	0.592 $\pm$ 0.014	0.588 $\pm$ 0.040	0.658 $\pm$ 0.092	0.552 $\pm$ 0.012	0.383 $\pm$ 0.014	0.527 $\pm$ 0.007	0.575 $\pm$ 0.010	0.505 $\pm$ 0.010	0.559 $\pm$ 0.008	0.562 $\pm$ 0.017	0.544 $\pm$ 0.014	0.662 $\pm$ 0.000
IForest	0.525 $\pm$ 0.080	0.507 $\pm$ 0.032	0.672 $\pm$ 0.004	0.000 $\pm$ 0.000	0.394 $\pm$ 0.011	0.546 $\pm$ 0.019	0.684 $\pm$ 0.003	0.568 $\pm$ 0.005	0.546 $\pm$ 0.012	0.519 $\pm$ 0.005	0.519 $\pm$ 0.002	0.607 $\pm$ 0.003
INNE	0.534 $\pm$ 0.095	0.616 $\pm$ 0.057	0.647 $\pm$ 0.050	0.697 $\pm$ 0.152	0.382 $\pm$ 0.002	0.535 $\pm$ 0.002	0.668 $\pm$ 0.008	0.549 $\pm$ 0.000	0.590 $\pm$ 0.008	0.557 $\pm$ 0.005	0.542 $\pm$ 0.000	0.734$\pm$0.010
KDE	0.446 $\pm$ 0.067	0.613 $\pm$ 0.028	0.476 $\pm$ 0.010	0.509 $\pm$ 0.128	0.406 $\pm$ 0.012	0.556 $\pm$ 0.000	0.650 $\pm$ 0.011	0.578 $\pm$ 0.020	0.589 $\pm$ 0.006	0.569 $\pm$ 0.004	0.551 $\pm$ 0.002	0.649 $\pm$ 0.007
KNN	0.461 $\pm$ 0.032	0.592 $\pm$ 0.074	0.654 $\pm$ 0.005	0.588 $\pm$ 0.014	0.412 $\pm$ 0.003	0.571 $\pm$ 0.002	0.548 $\pm$ 0.000	0.531 $\pm$ 0.010	0.562 $\pm$ 0.011	0.551 $\pm$ 0.004	0.537 $\pm$ 0.008	0.667 $\pm$ 0.005
KPCA	0.583 $\pm$ 0.020	0.567 $\pm$ 0.053	0.731 $\pm$ 0.019	0.000 $\pm$ 0.000	0.411 $\pm$ 0.004	0.577 $\pm$ 0.004	0.518 $\pm$ 0.004	0.516 $\pm$ 0.004	0.598 $\pm$ 0.015	0.559 $\pm$ 0.005	0.548 $\pm$ 0.001	0.714 $\pm$ 0.011
LMDD	0.388 $\pm$ 0.058	0.639$\pm$0.007	0.716 $\pm$ 0.013	0.634 $\pm$ 0.012	0.412 $\pm$ 0.008	0.566 $\pm$ 0.002	0.562 $\pm$ 0.009	0.520 $\pm$ 0.006	0.593 $\pm$ 0.008	0.520 $\pm$ 0.000	0.504 $\pm$ 0.004	0.642 $\pm$ 0.002
LODA	0.566 $\pm$ 0.045	0.569 $\pm$ 0.016	0.488 $\pm$ 0.017	0.000 $\pm$ 0.000	0.392 $\pm$ 0.001	0.544 $\pm$ 0.007	0.622 $\pm$ 0.010	0.491 $\pm$ 0.008	0.608 $\pm$ 0.006	0.532 $\pm$ 0.005	0.510 $\pm$ 0.001	0.684 $\pm$ 0.005
LOF	0.513 $\pm$ 0.055	0.554 $\pm$ 0.056	0.694 $\pm$ 0.000	0.000 $\pm$ 0.000	0.396 $\pm$ 0.002	0.497 $\pm$ 0.001	0.695 $\pm$ 0.017	0.507 $\pm$ 0.012	0.573 $\pm$ 0.002	0.524 $\pm$ 0.002	0.511 $\pm$ 0.011	0.708 $\pm$ 0.012
LSCP	0.526 $\pm$ 0.008	0.634 $\pm$ 0.008	0.577 $\pm$ 0.024	0.000 $\pm$ 0.000	0.404 $\pm$ 0.000	0.535 $\pm$ 0.018	0.497 $\pm$ 0.015	0.528 $\pm$ 0.013	0.581 $\pm$ 0.004	0.532 $\pm$ 0.001	0.529 $\pm$ 0.005	0.590 $\pm$ 0.001
LUNAR	0.500 $\pm$ 0.046	0.627 $\pm$ 0.031	0.558 $\pm$ 0.062	0.000 $\pm$ 0.000	0.394 $\pm$ 0.004	0.550 $\pm$ 0.013	0.500 $\pm$ 0.022	0.535 $\pm$ 0.032	0.557 $\pm$ 0.001	0.507 $\pm$ 0.012	0.488 $\pm$ 0.002	0.673 $\pm$ 0.001
MCD	0.438 $\pm$ 0.061	0.565 $\pm$ 0.009	0.726 $\pm$ 0.031	0.000 $\pm$ 0.000	0.420 $\pm$ 0.007	0.566 $\pm$ 0.036	0.500 $\pm$ 0.018	0.547 $\pm$ 0.025	0.579 $\pm$ 0.009	0.517 $\pm$ 0.005	0.514 $\pm$ 0.022	0.609 $\pm$ 0.003
OCSVM	0.532 $\pm$ 0.039	0.580 $\pm$ 0.024	0.648 $\pm$ 0.010	0.000 $\pm$ 0.000	0.383 $\pm$ 0.017	0.544 $\pm$ 0.013	0.688 $\pm$ 0.015	0.584 $\pm$ 0.002	0.587 $\pm$ 0.002	0.524 $\pm$ 0.010	0.523 $\pm$ 0.002	0.587 $\pm$ 0.000
PCA	0.461 $\pm$ 0.010	0.591 $\pm$ 0.028	0.776$\pm$0.032	0.000 $\pm$ 0.000	0.403 $\pm$ 0.002	0.583 $\pm$ 0.007	0.543 $\pm$ 0.004	0.493 $\pm$ 0.014	0.558 $\pm$ 0.006	0.532 $\pm$ 0.006	0.513 $\pm$ 0.006	0.634 $\pm$ 0.016
Sampling	0.604 $\pm$ 0.006	0.618 $\pm$ 0.012	0.561 $\pm$ 0.007	0.706 $\pm$ 0.014	0.372 $\pm$ 0.012	0.501 $\pm$ 0.018	0.738 $\pm$ 0.019	0.566 $\pm$ 0.006	0.595 $\pm$ 0.008	0.573 $\pm$ 0.006	0.564 $\pm$ 0.021	0.659 $\pm$ 0.002
SOD	0.581 $\pm$ 0.024	0.550 $\pm$ 0.066	0.538 $\pm$ 0.023	0.661 $\pm$ 0.121	0.392 $\pm$ 0.015	0.536 $\pm$ 0.016	0.708 $\pm$ 0.005	0.546 $\pm$ 0.003	0.620 $\pm$ 0.005	0.567 $\pm$ 0.008	0.544 $\pm$ 0.001	0.732 $\pm$ 0.001
SUOD	0.561 $\pm$ 0.011	0.599 $\pm$ 0.010	0.594 $\pm$ 0.009	0.453 $\pm$ 0.005	0.392 $\pm$ 0.013	0.564 $\pm$ 0.001	0.539 $\pm$ 0.005	0.560 $\pm$ 0.007	0.527 $\pm$ 0.009	0.552 $\pm$ 0.007	0.545 $\pm$ 0.006	0.694 $\pm$ 0.006
ALOE	0.412 $\pm$ 0.103	0.528 $\pm$ 0.013	0.251 $\pm$ 0.027	0.753$\pm$0.014	0.450$\pm$0.040	0.587$\pm$0.012	0.773$\pm$0.007	0.712$\pm$0.033	0.500 $\pm$ 0.032	0.891$\pm$0.046	0.621$\pm$0.023	0.464 $\pm$ 0.032

match exactly between the model prediction results and the actual set of labels in multidimensional anomaly detection. Specifically, for each sample, if all the labels predicted by the model are exactly the same as the actual labels, SEM of the sample is 1, otherwise it is 0, as shown in Table IV. SEM index of the whole dataset can be obtained by averaging the SEM values of all samples. When the value of SEM is high, it indicates that the model has good prediction performance in anomaly detection of heterogeneous label space, and can accurately predict the state of each sample label space. This means that the model has high precision and accuracy in state prediction, and can effectively identify the correlation between samples in different label spaces. A high SEM value usually means that the model has a strong anomaly detection ability in complex scenarios and can provide users with more accurate and reliable results.

We also use the average AUC to evaluate the performance of ALOE. As shown in Table V, ALOE achieves the highest average AUC on “Flare1,” “WQanimals,” “WQplants,” “Voice,” “Song,” “COIL2000,” and “TIC2000” datasets. This shows that ALOE can distinguish well between normal samples and outliers in each label space. However, we also notice that on “Edm,” “Jura,” “Enb,” and “Pain” datasets, our experimental results are not satisfactory. Through detailed analyses, we find that the “Enb” and “Pain” datasets have a lower AUC for ALOE on a certain label space, which lead to a decrease in the value of the average AUC as well. This is due to the fact that there are multiple types of outliers in this label space, and some of outliers are distributed on the same side as the normal samples, leading to the poor detection results of the one-class-based anomaly detection methods in this label space. It is worth noting that the output of ALOE is the label, not the probabilities of anomalies. When we convert the results

from labels to anomaly probabilities, if the samples have very close distance values, it may affect their ordering, leading to some differences in the results of AUC, as in the “Edm” and “BeLaE” datasets.

2) *Parameter Sensitivity Analysis*: We further conduct the parameter sensitivity test to investigate the relationship between the performance of ALOE and its parameters. We present the experimental results of four evaluation metrics when parameters λ_1 and λ_2 vary within the range $\{0.2, 0.4, \dots, 1.0\}$, as shown in Figs. 2–5. Regarding HS, SEM, and average AUC, we observe that across 12 datasets, the fluctuation of these metrics’ results remains minimal as the two parameters increase. One potential reason for this is that the normal samples and outliers in the datasets are highly distinguishable, enabling ALOE to consistently identify outliers in each label space regardless of parameter variations. Furthermore, this outcome demonstrates the high stability of ALOE, indicating that parameter changes within a certain range do not significantly impact its performance.

For EM, as shown in Fig. 3, we notice that the variation in EM is minimal on the datasets “Jura,” “Flare1,” “WQanimals,” “Voice,” “Song,” “BeLaE,” “COIL2000,” “TIC2000,” and “Pain.” This result aligns with our previous analysis, suggesting that the distributions of normal and abnormal classes in these datasets are relatively dispersed, rendering ALOE less sensitive to parameter changes. However, on “Edm” dataset, EM exhibits random fluctuations, albeit with minor overall changes. In contrast, EM results on the “WQplants” dataset gradually decrease as the parameters increase. We posit that when both balancing parameters are large, our method tends to select more support vectors, leading to a more complex decision boundary. Additionally, when the parameters are large, the method assigns a high weight to the recognition

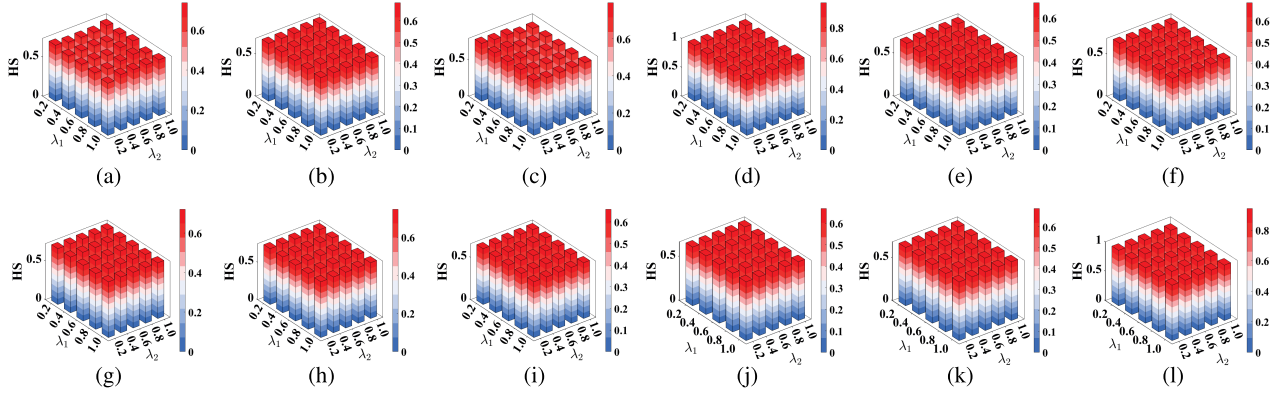


Fig. 2. HS on 12 datasets with different parameters. (a) Edm. (b) Jura. (c) Enb. (d) Flare1. (e) WQanimals. (f) WQplants. (g) Voice. (h) Song. (i) BeLaE. (j) COIL2000. (k) TIC2000. (l) Pain.

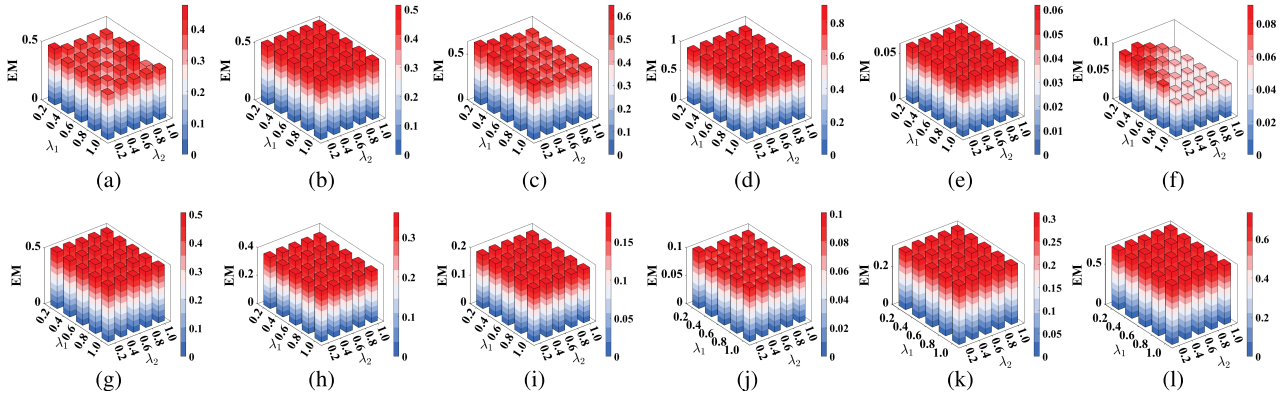


Fig. 3. EM on 12 datasets with different parameters. (a) Edm. (b) Jura. (c) Enb. (d) Flare1. (e) WQanimals. (f) WQplants. (g) Voice. (h) Song. (i) BeLaE. (j) COIL2000. (k) TIC2000. (l) Pain.

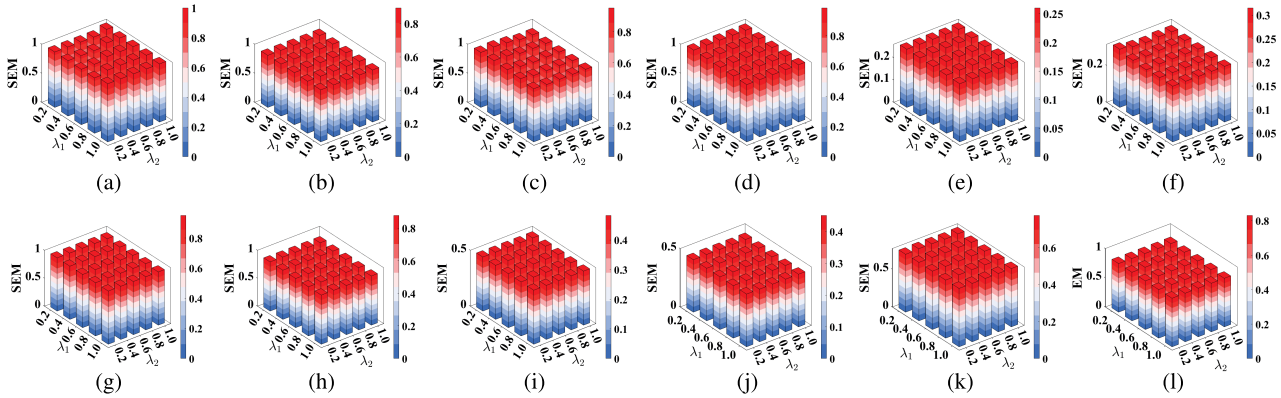


Fig. 4. SEM on 12 datasets with different parameters. (a) Edm. (b) Jura. (c) Enb. (d) Flare1. (e) WQanimals. (f) WQplants. (g) Voice. (h) Song. (i) BeLaE. (j) COIL2000. (k) TIC2000. (l) Pain.

errors of each training sample, making it more sensitive to noise in the dataset.

3) *Correlation Analysis*: We then analyze the correlation between the dimensions on each dataset, as shown in Fig. 6. Specifically, the red block indicates that the two dimensions are dependent, and a darker red block is a higher degree of dependence. This means that there is a potential homoscedastic influence between the two dimensions. When a sample in one dimension is abnormal, there is a certain probability that it will lead to an abnormality of this sample in the other dimension. On the contrary, the blue block can be interpreted

as the existence of a mutually exclusive relationship between the two dimensions, with a darker blue block indicating a higher degree of mutual exclusivity. This implies that there is a potential reverse influence between the two dimensions. When a sample is outlier in one dimension, there is a high probability that it will be concluded that the samples in the other dimension are behaving normally. White block denotes that the two dimensions are independent of each other. The abnormality of the sample in one dimension has no effect on whether the sample in the other dimension are abnormal or not. As shown in Fig. 6, there are dependencies between

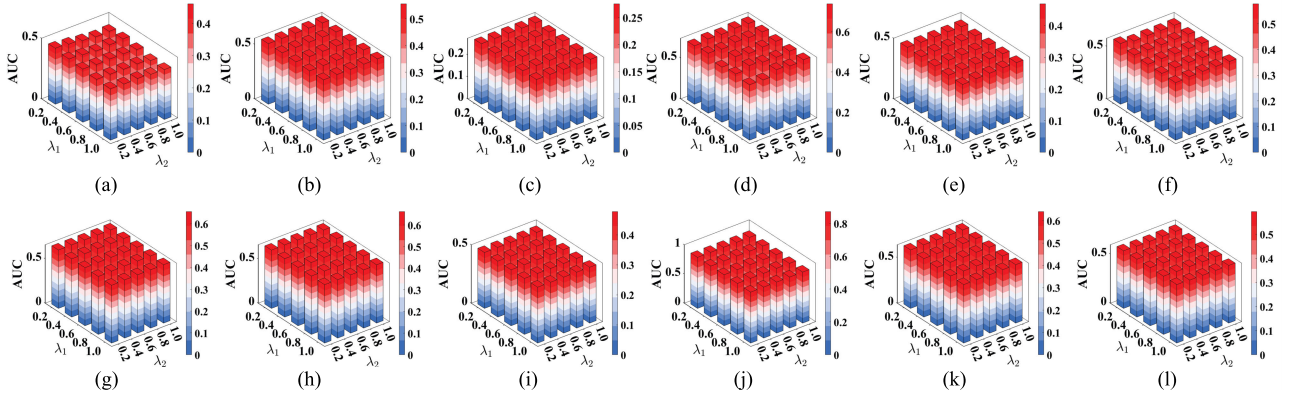


Fig. 5. Average AUC on 12 datasets with different parameters. (a) Edm. (b) Jura. (c) Enb. (d) Flare1. (e) WQanimals. (f) WQplants. (g) Voice. (h) Song. (i) BeLaE. (j) COIL2000. (k) TIC2000. (l) Pain.

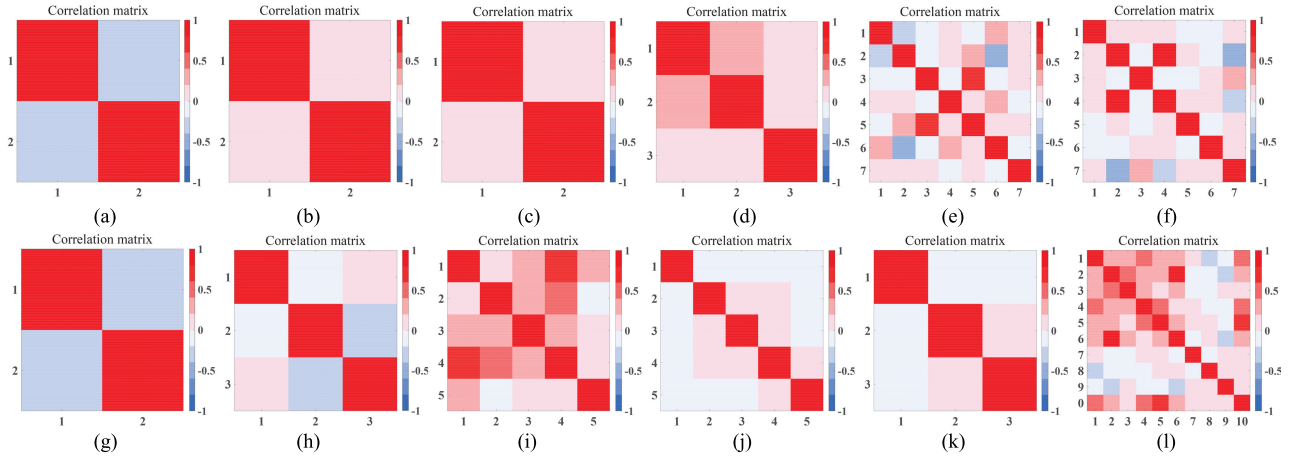


Fig. 6. Correlation matrix results on 12 datasets (The coordinates represent the dimension of the dataset). (a) Edm. (b) Jura. (c) Enb. (d) Flare1. (e) WQanimals. (f) WQplants. (g) Voice. (h) Song. (i) BeLaE. (j) COIL2000. (k) TIC2000. (l) Pain.

several dimensions on the datasets “Jura,” “Enb,” “Flare1,” “BeLaE,” “COIL2000,” and “TIC2000.” In contrast, on the datasets “Edm” and “Voice,” there are both dependencies and mutual exclusions between dimensions. And for the datasets “WQanimals,” “WQplants,” “Song,” and “Pain,” there are all three of these relationships between two dimensions. The captured dependencies between two dimensions help to generate the anomaly detection method with strong generalization performance, which is the key to enable ALOE to effectively deal with the multidimensional anomaly detection.

V. CONCLUSION

In this article, we attempt to study the multidimensional anomaly detection for the first time. For a given dataset, this task aims to identify outliers that significantly deviate from normal distribution in different dimensions with specific semantic information. However, the heterogeneity of the dimension poses a severe challenge for this specific anomaly detection, as it results in the incomparability of modeling outputs across different dimensions. This article introduces maximum margin technique and covariance regularization for multidimensional anomaly detection and proposes ALOE. ALOE constructs a convex optimization problem incorporating nonlinear constraints. The objective function is to learn a decision boundary that can distinguish between normal samples and outliers in the same dimension through the maximum

margin, and to capture the relationships among different dimensions through covariance regularization. To obtain the optimal decision boundary under each dimension, we design the alternating optimization method for this convex optimization problem. In the future, we try to explore multidimensional anomaly detection under multiple abnormal types.

APPENDIX

A. Partial Derivatives of $\mathbf{W}$, ρ , and ξ for the Lagrangian Function

Let these results be equal to zero. We can get the following equations:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial \mathbf{W}} = 0 \Rightarrow \mathbf{W} = \sum_{j=1}^m \sum_{i=1}^{n_j} \alpha_i^j \mathbf{x}_i^j \mathbf{e}_j^\top \mathbf{C} (\lambda_1 \mathbf{C} + \lambda_2 \mathbf{I}_m)^{-1} \\ \quad + \sum_{j=1}^m \sum_{l=n_j+1}^n \alpha_l^j \mathbf{x}_l^j \mathbf{e}_j^\top \mathbf{C} (\lambda_1 \mathbf{C} + \lambda_2 \mathbf{I}_m)^{-1} \\ \frac{\partial \mathcal{L}}{\partial \rho_j} = 0 \Rightarrow \sum_{i=1}^{n_j} \alpha_i^j + \sum_{l=n_j+1}^n \alpha_l^j = 1 \\ \frac{\partial \mathcal{L}}{\partial \xi_i^j} = 0 \Rightarrow \alpha_i^j + \beta_i^j = 1, \quad \frac{\partial \mathcal{L}}{\partial \xi_l^j} = 0 \Rightarrow \alpha_l^j + \beta_l^j = 1 \end{cases}$$

where $\mathbf{e}_j$ denotes the j th column of identity matrix $\mathbf{I}_m$.

B. Proof of Lemma 1

Proof: Let $(\mathbf{W}^\top \mathbf{W})^{(1/2)} \mathbf{C}^{-(1/2)} = \mathbf{U}$, $\mathbf{C}^{(1/2)} = \mathbf{V}$. According to the property of matrix,

$$\begin{aligned} \text{tr}(\mathbf{U}^\top \mathbf{U}) &= \sum_{i,j=1}^m \mathbf{U}_{ij}^2 = \langle \mathbf{U}_{\text{vec}}, \mathbf{U}_{\text{vec}} \rangle = \|\mathbf{U}_{\text{vec}}\|^2 \\ \text{tr}(\mathbf{V}^\top \mathbf{V}) &= \sum_{i,j=1}^m \mathbf{V}_{ij}^2 = \langle \mathbf{V}_{\text{vec}}, \mathbf{V}_{\text{vec}} \rangle = \|\mathbf{V}_{\text{vec}}\|^2 \\ \text{tr}(\mathbf{U}^\top \mathbf{V}) &= \sum_{i,j=1}^m \mathbf{U}_{ij} \mathbf{V}_{ij} = \langle \mathbf{U}_{\text{vec}}, \mathbf{V}_{\text{vec}} \rangle \end{aligned}$$

where $\mathbf{U}_{\text{vec}}$ and $\mathbf{V}_{\text{vec}}$ are the vectorized form of $\mathbf{U}$ and $\mathbf{V}$, respectively. Based on Cauchy–Schwarz inequality, we have $\|\mathbf{U}_{\text{vec}}\| \cdot \|\mathbf{V}_{\text{vec}}\| \geq |\langle \mathbf{U}_{\text{vec}}, \mathbf{V}_{\text{vec}} \rangle|$. By squaring both sides of this inequality, we have $\|\mathbf{U}_{\text{vec}}\|^2 \cdot \|\mathbf{V}_{\text{vec}}\|^2 \geq (\langle \mathbf{U}_{\text{vec}}, \mathbf{V}_{\text{vec}} \rangle)^2$. The equality relationship holds only when $\mathbf{U}_{\text{vec}} = \mu \cdot \mathbf{V}_{\text{vec}}$, i.e., $\mathbf{U} = \mu \cdot \mathbf{V}$. ■

C. Kernel Extension for ALOE

We further give the nonlinear version of ALOE for improving the condition that the data distribution is complicated. The optimization problem is as follows:

$$\begin{aligned} \min_{\mathbf{w}, \rho, \xi, \mathbf{C}} \quad & \frac{\lambda_1}{2} \text{tr}(\mathbf{W}\mathbf{W}^\top) + \frac{\lambda_2}{2} \text{tr}(\mathbf{W}\mathbf{C}^{-1}\mathbf{W}^\top) - \sum_{j=1}^m \rho_j \\ & + \sum_{j=1}^m \sum_{i=1}^{n_j} \xi_i^j + \sum_{j=1}^m \sum_{l=n_j+1}^n \xi_l^j \\ \text{s.t.} \quad & \mathbf{C} \succeq 0, \quad \text{tr}(\mathbf{C}) \leq 1 \\ & \mathbf{w}_j^\top \phi(\mathbf{x}_i^j) \geq \rho_j - \xi_i^j, \quad i = 1, \dots, n_j \\ & \mathbf{w}_j^\top \phi(\mathbf{x}_l^j) \leq \rho_j + \xi_l^j, \quad l = n_j + 1, \dots, n \\ & \xi_i^j, \xi_l^j \geq 0, \quad j = 1, \dots, m. \end{aligned} \quad (11)$$

The solution method is the same as above. Specifically, when we fix $\mathbf{C}$, the optimization problem (11) is transformed as

$$\begin{aligned} \min_{\mathbf{w}, \rho, \xi} \quad & \frac{\lambda_1}{2} \text{tr}(\mathbf{W}\mathbf{W}^\top) + \frac{\lambda_2}{2} \text{tr}(\mathbf{W}\mathbf{C}^{-1}\mathbf{W}^\top) - \sum_{j=1}^m \rho_j \\ & + \sum_{j=1}^m \sum_{i=1}^{n_j} \xi_i^j + \sum_{j=1}^m \sum_{l=n_j+1}^n \xi_l^j \\ \text{s.t.} \quad & \mathbf{w}_j^\top \phi(\mathbf{x}_i^j) \geq \rho_j - \xi_i^j, \quad i = 1, \dots, n_j \\ & \mathbf{w}_j^\top \phi(\mathbf{x}_l^j) \leq \rho_j + \xi_l^j, \quad l = n_j + 1, \dots, n \\ & \xi_i^j, \xi_l^j \geq 0, \quad j = 1, \dots, m. \end{aligned}$$

The dual problem of the above optimization problem is as follows:

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2} \sum_{j_1, j_2=1}^m \sum_{i_1=1}^{n_{j_1}} \sum_{i_2=1}^{n_{j_2}} \alpha_{i_1}^{j_1} \alpha_{i_2}^{j_2} \mathbf{H}_{j_1 j_2} \langle \phi(\mathbf{x}_{i_1}^{j_1}), \phi(\mathbf{x}_{i_2}^{j_2}) \rangle \\ & + \sum_{j_1, j_2=1}^m \sum_{l_1=n_{j_1}+1}^n \sum_{l_2=n_{j_2}+1}^n \alpha_{l_1}^{j_1} \alpha_{l_2}^{j_2} \mathbf{H}_{j_1 j_2} \langle \phi(\mathbf{x}_{l_1}^{j_1}), \phi(\mathbf{x}_{l_2}^{j_2}) \rangle \end{aligned}$$

$$\begin{aligned} \text{s.t.} \quad & \sum_{i=1}^{n_j} \alpha_i^j + \sum_{l=n_j+1}^n \alpha_l^j = 1 \\ & 0 \leq \alpha_i^j \leq 1, \quad 0 \leq \alpha_l^j \leq 1, \quad j = 1, \dots, m. \end{aligned}$$

We introduce the kernel trick with a kernel function $\kappa(\mathbf{x}_{i_1}^{j_1}, \mathbf{x}_{i_2}^{j_2}) = \langle \phi(\mathbf{x}_{i_1}^{j_1}), \phi(\mathbf{x}_{i_2}^{j_2}) \rangle$, the dual problem can be rewritten as follows:

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2} \sum_{j_1, j_2=1}^m \sum_{i_1=1}^{n_{j_1}} \sum_{i_2=1}^{n_{j_2}} \alpha_{i_1}^{j_1} \alpha_{i_2}^{j_2} \mathbf{H}_{j_1 j_2} \kappa(\mathbf{x}_{i_1}^{j_1}, \mathbf{x}_{i_2}^{j_2}) \\ & + \sum_{j_1, j_2=1}^m \sum_{l_1=n_{j_1}+1}^n \sum_{l_2=n_{j_2}+1}^n \alpha_{l_1}^{j_1} \alpha_{l_2}^{j_2} \mathbf{H}_{j_1 j_2} \kappa(\mathbf{x}_{l_1}^{j_1}, \mathbf{x}_{l_2}^{j_2}) \\ \text{s.t.} \quad & \sum_{i=1}^{n_j} \alpha_i^j + \sum_{l=n_j+1}^n \alpha_l^j = 1 \\ & 0 \leq \alpha_i^j \leq 1, \quad 0 \leq \alpha_l^j \leq 1, \quad j = 1, \dots, m. \end{aligned}$$

REFERENCES

- [1] X. Wang, L. Duan, Z. Yu, C. He, and Z. Bao, "Robust multi-kernel nearest neighborhood for outlier detection," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 8, pp. 4220–4231, Aug. 2024.
- [2] S. Dzeroski, D. Demsar, and J. Grbovic, "Predicting chemical parameters of river water quality from bioindicator data," *Appl. Intell.*, vol. 13, no. 1, pp. 7–17, 2000.
- [3] R. Walecki, O. Rudovic, V. Pavlovic, and M. Pantic, "Copula ordinal regression framework for joint estimation of facial action unit intensity," *IEEE Trans. Affect. Comput.*, vol. 10, no. 3, pp. 297–312, Jul. 2019.
- [4] B.-B. Jia and M.-L. Zhang, "Multi-dimensional classification via stacked dependency exploitation," *Sci. China Inf. Sci.*, vol. 63, no. 12, p. 222102, 2020.
- [5] C. C. Aggarwal, *Outlier Analysis*. Berlin, Germany: Springer, 2013.
- [6] A. Boukerche, L. Zheng, and O. Alfandi, "Outlier detection: Methods, models, and classification," *ACM Comput. Surv.*, vol. 53, no. 3, p. 55, 2021.
- [7] G. Pang, C. Shen, L. Cao, and A. van den Hengel, "Deep learning for anomaly detection: A review," *ACM Comput. Surv.*, vol. 54, no. 2, p. 38, 2022.
- [8] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, Jul. 2001.
- [9] Y. Song, Z. Wen, C. Lin, and R. Davis, "One-class conditional random fields for sequential anomaly detection," in *Proc. 23rd Int. Joint Conf. Artif. Intell. (IJCAI)*, 2013, pp. 1685–1691.
- [10] M. Swarnkar and N. Hubballi, "OCPAD: One class naive Bayes classifier for payload based anomaly detection," *Expert Syst. Appl.*, vol. 64, pp. 330–339, Dec. 2016.
- [11] Y. Tian, M. Mirzabagheri, S. M. H. Bamakan, H. Wang, and Q. Qu, "Ramp loss one-class support vector machine; A robust and effective approach to anomaly detection problems," *Neurocomputing*, vol. 310, pp. 223–235, Oct. 2018.
- [12] C. Gautam, R. Balaji, K. Sudharsan, A. Tiwari, and K. Ahuja, "Localized multiple kernel learning for anomaly detection: One-class classification," *Knowl.-Based Syst.*, vol. 165, pp. 241–252, Feb. 2019.
- [13] X. Miao, Y. Liu, H. Zhao, and C. Li, "Distributed online one-class support vector machine for anomaly detection over networks," *IEEE Trans. Cybern.*, vol. 49, no. 4, pp. 1475–1488, Apr. 2019.
- [14] J. He, Z. Cheng, and B. Guo, "Anomaly detection in telemetry data using a jointly optimal one-class support vector machine with dictionary learning," *Rel. Eng. Syst. Saf.*, vol. 242, Feb. 2024, Art. no. 109717.
- [15] D. M. J. Tax and R. P. W. Duin, "Support vector data description," *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, Jan. 2004.
- [16] E. J. Pauwels and O. Ambekar, "One class classification for anomaly detection: Support vector data description revisited," in *Proc. 11th Ind. Conf. Adv. Data Mining Appl. Theor. Aspects (ICDM)* (Lecture Notes in Computer Science), vol. 6870. New York, NY, USA: Springer, 2011, pp. 25–39.
- [17] J. Wang and A. Cherian, "GODS: Generalized one-class discriminative subspaces for anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8200–8210.

- [18] X. Wang, L. Duan, C. He, Y. Chen, and X. Wu, "An efficient adaptive multi-kernel learning with safe screening rule for outlier detection," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 8, pp. 3656–3669, Aug. 2024.
- [19] H. Wu and Y.-F. Li, "Adaptive graph-based support vector data description for weakly-supervised anomaly detection," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 3, pp. 3863–3872, Jun. 2024.
- [20] L. Ruff et al., "Deep one-class classification," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 4390–4399.
- [21] S. M. Erfani, S. Rajasegarar, S. Karunasekera, and C. Leckie, "High-dimensional and large-scale anomaly detection using a linear one-class SVM with deep learning," *Pattern Recognit.*, vol. 58, pp. 121–134, Oct. 2016.
- [22] L. Shen, Z. Li, and J. T. Kwok, "Timeseries anomaly detection using temporal hierarchical one-class network," in *Proc. Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 1–11.
- [23] S. Goyal, A. Raghunathan, M. Jain, H. V. Simhadri, and P. Jain, "DROCC: Deep robust one-class classification," in *Proc. 37th Int. Conf. Mach. Learn.*, 2020, pp. 3711–3721.
- [24] Y. Chen, Y. Tian, G. Pang, and G. Carneiro, "Deep one-class classification via interpolated Gaussian descriptor," in *Proc. 36th AAAI Conf. Artif. Intell. (AAAI)*, 2022, pp. 383–392.
- [25] L. Maziarka, M. Smieja, M. Sendera, L. Struski, J. Tabor, and P. Spurek, "OneFlow: One-class flow for anomaly detection based on a minimal volume region," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8508–8519, Nov. 2022.
- [26] F. V. Massoli, F. Falchi, A. Kantarci, S. Akti, H. K. Ekenel, and G. Amato, "MOCCA: Multilayer one-class classification for anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 6, pp. 2313–2323, Jun. 2022.
- [27] H. Deng and X. Li, "Anomaly detection via reverse distillation from one-class embedding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9727–9736.
- [28] Y. Qiao, K. Wu, and P. Jin, "Efficient anomaly detection for high-dimensional sensing data with one-class support vector machine," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 404–417, Jan. 2023.
- [29] N. Belton, M. T. Hagos, A. Lawlor, and K. M. Curran, "FewSOME: One-class few shot anomaly detection with Siamese networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 2978–2987.
- [30] M. Yang, J. Liu, Z. Yang, and Z. Wu, "SLSG: Industrial image anomaly detection with improved feature embeddings and one-class classification," *Pattern Recognit.*, vol. 156, Dec. 2024, Art. no. 110862.
- [31] J. Chi and Z. Mao, "Deep domain-adversarial anomaly detection with robust one-class transfer learning," *Knowl.-Based Syst.*, vol. 300, Sep. 2024, Art. no. 112225.
- [32] B.-B. Jia and M.-L. Zhang, "Multi-dimensional classification via decomposed label encoding," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 2, pp. 1844–1856, Feb. 2023.
- [33] J. Read, C. Bielza, and P. Larrañaga, "Multi-dimensional classification with super-classes," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 7, pp. 1720–1733, Jul. 2014.
- [34] H. Wang, C. Chen, W. Liu, K. Chen, T. Hu, and G. Chen, "Incorporating label embedding and feature augmentation for multi-dimensional classification," in *Proc. 34th AAAI Conf. Artif. Intell. (AAAI)*, 2020, pp. 6178–6185.
- [35] B.-B. Jia and M.-L. Zhang, "Multi-dimensional classification via kNN feature augmentation," *Pattern Recognit.*, vol. 106, Oct. 2020, Art. no. 107423.
- [36] B. Jia and M. Zhang, "Multi-dimensional classification via sparse label encoding," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, vol. 139, 2021, pp. 4917–4926.
- [37] B.-B. Jia and M.-L. Zhang, "Maximum margin multi-dimensional classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7185–7198, Dec. 2022.
- [38] T. Huang, B. Jia, and M. Zhang, "Progressive label propagation for semi-supervised multi-dimensional classification," in *Proc. 32nd Int. Joint Conf. Artif. Intell. (IJCAI)*, 2023, pp. 3821–3829.
- [39] B.-B. Jia and M.-L. Zhang, "Multi-dimensional multi-label classification: Towards encompassing heterogeneous label spaces and multi-label annotations," *Pattern Recognit.*, vol. 138, Jun. 2023, Art. no. 109357.
- [40] X. Liu, J. Zhu, Z. Tian, and Z. Li, "Multi-dimensional classification via class space fusion and comprehensive label correlations," *Inf. Fusion*, vol. 111, Nov. 2024, Art. no. 102521.
- [41] Z. You et al., "A unified model for multi-class anomaly detection," in *Proc. 35th Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 1–14.
- [42] R. Lu et al., "Hierarchical vector quantized transformer for multi-class unsupervised anomaly detection," in *Proc. 36th Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2023, pp. 1–14.
- [43] C. Wang and A. Tavakkoli, "Latent diffusion based multi-class anomaly detection," in *Proc. 18th Int. Symp. Vis. Comput. (ISVC)*, vol. 14361, 2023, pp. 487–498.
- [44] Y. Li et al., "Vague prototype-oriented diffusion model for multi-class anomaly detection," in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, 2024, pp. 1–20.
- [45] H. He et al., "A diffusion-based framework for multi-class anomaly detection," in *Proc. 38th AAAI Conf. Artif. Intell. (AAAI)*, 2024, pp. 8472–8480.
- [46] J. Liu and F. Wang, "Mixed-attention auto encoder for multi-class industrial anomaly detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 4120–4124.
- [47] C. Wu, X. Liu, J. Wu, H. Zhang, and L. Wang, "Vertical-horizontal latent space with iterative memory review network for multi-class anomaly detection," *Knowl.-Based Syst.*, vol. 292, May 2024, Art. no. 111594.
- [48] B. Kang, Y. Zhong, Z. Sun, L. Deng, M. Wang, and J. Zhang, "MSTAD: A masked subspace-like transformer for multi-class anomaly detection," *Knowl.-Based Syst.*, vol. 283, Jan. 2024, Art. no. 111186.
- [49] E. Spyromitros-Xioufis, G. Tsoumakas, W. Groves, and I. Vlahavas, "Multi-target regression via input space expansion: Treating targets as inputs," *Mach. Learn.*, vol. 104, no. 1, pp. 55–98, 2016.
- [50] M. A. Álvarez and N. D. Lawrence, "Computationally efficient convolved multiple output Gaussian processes," *J. Mach. Learn. Res.*, vol. 12, no. 41, pp. 1459–1500, Feb. 2011.
- [51] A. Tsanas and A. Xifara, "Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools," *Energy Buildings*, vol. 49, pp. 560–567, Jun. 2012.
- [52] C. Liu, P. Zhao, S. Huang, Y. Jiang, and Z. Zhou, "Dual set multi-label learning," in *Proc. 32nd AAAI Conf. Artif. Intell. (AAAI)*, 2018, pp. 3635–3642.
- [53] C. Brinker, E. L. Mencía, and J. Fürnkranz, "Graded multilabel classification by pairwise comparisons," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*. Shenzhen, China: IEEE Computer Society, Dec. 2014, pp. 731–736.
- [54] P. Lucey, J. F. Cohn, K. M. Prkachin, P. E. Solomon, S. Chew, and I. Matthews, "Painful monitoring: Automatic pain monitoring using the UNBC-McMaster shoulder pain expression archive database," *Image Vis. Comput.*, vol. 30, no. 3, pp. 197–205, Mar. 2012.
- [55] Z. He, X. Xu, and S. Deng, "Discovering cluster-based local outliers," *Pattern Recognit. Lett.*, vol. 24, nos. 9–10, pp. 1641–1650, 2003.
- [56] J. Tang, Z. Chen, A. W. Fu, and D. W. Cheung, "Enhancing effectiveness of outlier detections for low density patterns," in *Proc. Pacific-Asia Conf. Adv. Knowl. Discovery Data Mining (PAKDD)*, 2002, pp. 535–548.
- [57] Z. Li, Y. Zhao, N. Botta, C. Ionescu, and X. Hu, "COPOD: Copula-based outlier detection," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2020, pp. 1118–1123.
- [58] Z. Li, Y. Zhao, X. Hu, N. Botta, C. Ionescu, and G. Chen, "ECOD: Unsupervised outlier detection using empirical cumulative distribution functions," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 12, pp. 12181–12193, Mar. 2022.
- [59] A. Lazarevic and V. Kumar, "Feature bagging for outlier detection," in *Proc. 11th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2005, pp. 157–166.
- [60] Y. Zhao, Z. Nasrullah, and Z. Li, "PyOD: A Python toolbox for scalable outlier detection," *J. Mach. Learn. Res.*, vol. 20, no. 96, pp. 1–7, 2019.
- [61] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Min.*, 2008, pp. 413–422.
- [62] T. R. Bandaragoda, K. M. Ting, D. Albrecht, F. T. Liu, Y. Zhu, and J. R. Wells, "Isolation-based anomaly detection using nearest-neighbor ensembles," *Comput. Intell.*, vol. 34, no. 4, pp. 968–998, Nov. 2018.
- [63] H. Hoffmann, "Kernel PCA for novelty detection," *Pattern Recognit.*, vol. 40, no. 3, pp. 863–874, 2007.
- [64] S. Ramaswamy, R. Rastogi, and K. Shim, "Efficient algorithms for mining outliers from large data sets," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, May 2000, pp. 427–438.
- [65] A. Arning, R. Agrawal, and P. Raghavan, "A linear method for deviation detection in large databases," in *Proc. Int. Conf. Knowl. Discovery Data Mining (SIGKDD)*, 1996, pp. 164–169.

- [66] T. Pevný, “Loda: Lightweight on-line detector of anomalies,” *Mach. Learn.*, vol. 102, no. 2, pp. 275–304, Feb. 2016.
- [67] M. M. Breunig, H. Kriegel, R. T. Ng, and J. Sander, “LOF: Identifying density-based local outliers,” in *Proc. ACM SIGMOD Int. Conf. Manage. Data (SIGMOD)*, 2000, pp. 93–104.
- [68] Y. Zhao, Z. Nasrullah, M. K. Hryniewicz, and Z. Li, “LSCP: Locally selective combination in parallel outlier ensembles,” in *Proc. SIAM Int. Conf. Data Mining*, 2019, pp. 585–593.
- [69] B. Schölkopf, R. C. Williamson, A. J. Smola, J. Shawe-Taylor, and J. C. Platt, “Support vector method for novelty detection,” in *Proc. Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 1999, pp. 582–588.
- [70] C. C. Aggarwal and P. S. Yu, “Outlier detection for high dimensional data,” in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, May 2001, pp. 37–46.
- [71] M. Sugiyama and K. M. Borgwardt, “Rapid distance-based outlier detection via sampling,” in *Proc. Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2013, pp. 467–475.
- [72] A. Agrawal, “Local subspace based outlier detection,” in *Proc. 2nd Int. Conf. Contemp. Comput. (IC3)*, vol. 40, 2009, pp. 149–157.
- [73] Y. Zhao et al., “SUOD: Accelerating large-scale unsupervised heterogeneous outlier detection,” in *Proc. Mach. Learn. Syst. (MLSys)*, 2021, pp. 1–16.
- [74] H. Xu, G. Pang, Y. Wang, and Y. Wang, “Deep isolation forest for anomaly detection,” *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 12, pp. 12591–12604, Dec. 2023.
- [75] H. Xu, Y. Wang, J. Wei, S. Jian, Y. Li, and N. Liu, “Fascinating supervisory signals and where to find them: Deep anomaly detection with scale learning,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 38655–38673.
- [76] T. Shenkar and L. Wolf, “Anomaly detection for tabular data with internal contrastive learning,” in *Proc. 10th Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–26.
- [77] B. Liu, D. Wang, K. Lin, P. Tan, and J. Zhou, “RCA: A deep collaborative autoencoder approach for anomaly detection,” in *Proc. 30th Int. Joint Conf. Artif. Intell. (IJCAI)*, 2021, pp. 1505–1511.
- [78] H. Wang, G. Pang, C. Shen, and C. Ma, “Unsupervised representation learning by predicting random distances,” in *Proc. 29th Int. Joint Conf. Artif. Intell. (IJCAI)*, 2020, pp. 2950–2956.
- [79] L. Bergman and Y. Hoshen, “Classification-based anomaly detection for general data,” in *Proc. 8th Int. Conf. Learn. Represent. (ICLR)*, 2020, pp. 1–12.
- [80] C. Qiu, T. Pfommer, M. Kloft, S. Mandt, and M. Rudolph, “Neural transformation learning for deep anomaly detection beyond images,” in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, vol. 139, 2021, pp. 8703–8714.
- [81] G. Pang, L. Cao, L. Chen, and H. Liu, “Learning representations of ultrahigh-dimensional data for random distance-based outlier detection,” in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2018, pp. 2041–2050.
- [82] H. Zenati, M. Romain, C. Foo, B. Lecouat, and V. Chandrasekhar, “Adversarially learned anomaly detection,” in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2018, pp. 727–736.
- [83] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014, pp. 1–14.
- [84] A. Goodge, B. Hooi, S. Ng, and W. S. Ng, “LUNAR: Unifying local outlier detection methods via graph neural networks,” in *Proc. Conf. Artif. Intell. (AAAI)*, 2022, pp. 6737–6745.



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